

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250280690

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

AOYAGI; KENICHI et al.

LIGHT EMITTING DEVICE, ELECTRONIC DEVICE, AND METHOD OF MANUFACTURING LIGHT EMITTING DEVICE

Abstract

Provided is a light emitting device that can facilitate a manufacturing process of a light emitting device having a plurality of emission colors and can suppress an increase in resistance of a second electrode, deterioration of optical characteristics, and luminance unevenness. A light emitting device includes: a plurality of sub-pixels that is two-dimensionally arranged and respectively corresponds to a plurality of emission colors; a connection portion that connects the plurality of different sub-pixels; a first electrode; and on an upper side of the first electrode, an organic layer including a light emitting layer, and a second electrode in this order, in which the first electrode and the organic layer are formed in at least each of the plurality of sub-pixels, the second electrode is formed in each of the plurality of sub-pixels and the connection portion, the connection portion is, in a case where a region between the plurality of sub-pixels is defined as an inter-sub-pixel region, formed in a part of the inter-sub-pixel region, and at least a part of the connection portion connects the plurality of sub-pixels having different emission colors.

Inventors: AOYAGI; KENICHI (KANAGAWA, JP), SHIMAYAMA; TSUTOMU (KANAGAWA, JP), ODA; NAOTO (KANAGAWA, JP)

Applicant: SONY SEMICONDUCTOR SOLUTIONS CORPORATION (KANAGAWA, JP)

Family ID: 88730405

Appl. No.: 18/862698

Filed (or PCT Filed): May 12, 2023

PCT No.: PCT/JP2023/017975

Foreign Application Priority Data

JP 2022-079026

May. 12, 2022

Publication Classification

Int. Cl.: H10K59/35 (20230101); H10K71/16 (20230101); H10K71/60 (20230101)

U.S. Cl.:

CPC H10K59/353 (20230201); H10K59/352 (20230201); H10K71/166 (20230201);
H10K71/60 (20230201);

Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a light emitting device, an electronic device, and a method of manufacturing a light emitting device.

BACKGROUND ART

[0002] As a light emitting device using a light emitting element such as an organic EL element, a display device having a structure in which an organic compound layer including a light emitting layer and a second electrode are layered on a first electrode formed in an arrangement pattern spaced apart from each other in units of sub-pixels constituting one pixel is known. Each layer such as the organic compound layer and the second electrode is pattern-formed using a forming method such as a method by a vapor deposition technique, a method by a printing technique, a method by etching, or a method using photolithography.

[0003] Regarding a display device as one of light emitting devices, Patent Document 1 discloses a technique in which a plurality of sub-pixels each including an organic compound layer corresponding to each of a plurality of emission colors is provided for each emission color, and the organic compound layer and the second electrode are pattern-formed such that the organic compound layer and the second electrode are connected between separated sub-pixels having the same emission color.

CITATION LIST

Patent Document

[0004] Patent Document 1: Japanese Patent Application Laid-Open No. 2021-163599

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

[0005] In Patent Document 1, in a case where the display device has the plurality of emission colors, a portion (referred to as an overlapping portion) in which a plurality of the second electrodes and a plurality of the organic compound layers are stacked may occur. In Patent Document 1, a manufacturing process is used in which the second electrodes constituting sub-pixels corresponding to different emission colors are separated from each other in the overlapping portion. In the manufacturing process, each of the second electrodes is formed for each emission color of the sub-pixel. Therefore, in Patent Document 1, there is room for improvement in terms of facilitation of the manufacturing process.

[0006] Moreover, in Patent Document 1, in a case where the display device has the plurality of emission colors, a step (unevenness) is generated at the overlapping portion. Therefore, in Patent Document 1, there is room for improvement in terms of suppressing a step due to the overlapping portion, suppressing an increase in resistance and deterioration of optical characteristics of the second electrode in a portion where the step is generated, and suppressing occurrence of luminance unevenness due to the increase in resistance of the second electrode.

[0007] The present disclosure has been made in view of the above-described points, and an object

of the present disclosure is to provide a light emitting device, an electronic device, and a method of manufacturing a light emitting device that can facilitate a manufacturing process of a light emitting device having a plurality of emission colors and can suppress an increase in resistance of a second electrode, deterioration of optical characteristics, and luminance unevenness.

Solutions to Problems

[0008] The present disclosure is, for example, (1) a light emitting device including: [0009] a plurality of sub-pixels that is two-dimensionally arranged and respectively corresponds to a plurality of emission colors; [0010] a connection portion that connects the plurality of different sub-pixels; [0011] a first electrode; and [0012] on an upper side of the first electrode, an organic layer including a light emitting layer, and a second electrode in this order, [0013] in which the first electrode and the organic layer are formed in at least each of the plurality of sub-pixels, [0014] the second electrode is formed in each of the plurality of sub-pixels and the connection portion, [0015] the connection portion is, in a case where a region between the plurality of sub-pixels is defined as an inter-sub-pixel region, formed in a part of the inter-sub-pixel region, and [0016] at least a part of the connection portion connects the plurality of sub-pixels having different emission colors.

[0017] The present disclosure may be (2) an electronic device including the display device according to (1) described above.

[0018] Furthermore, the present disclosure may be, for example, a method of manufacturing a light emitting device, the method including: [0019] a first step of patterning an organic layer including a light emitting layer on a first electrode using a mask determined according to a layout of a plurality of sub-pixels; [0020] a second step of layering a second electrode on the organic layer; and [0021] a third step of, using etching, removing a portion of the organic layer and the second electrode, the portion being out of a combined portion of the sub-pixels and a connection portion that connects the plurality of different sub-pixels.

[0022] Furthermore, the present disclosure is, for example, a method of manufacturing a light emitting device, the method including the steps of: [0023] forming a first organic layer including a first light emitting layer on a first electrode that forms two first sub-pixels adjacent to each other in a predetermined direction via a first mask; [0024] after disposing a second mask such that a first electrode that forms one third sub-pixel is present between an opening of the first mask and an opening of the second mask, forming a second organic layer including a second light emitting layer on a first electrode that forms two second sub-pixels adjacent to each other in the predetermined direction via the second mask; [0025] sequentially forming a second electrode that forms a first sub-pixel and a second sub-pixel, and a first protective layer on the first organic layer and the second organic layer; and [0026] forming the first sub-pixel and the second sub-pixel by patterning the first protective layer, the second electrode, the first organic layer, and the second organic layer such that a connection portion that connects the first sub-pixel and the second sub-pixel remains in an inter-sub-pixel region and the first electrode that forms the third sub-pixel is exposed.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0027] FIG. 1 is a plan view for explaining an example of a display device according to a first embodiment.

[0028] FIG. 2 is a partially enlarged plan view in which a portion of a region XS surrounded by a broken line in FIG. 1 is enlarged.

[0029] FIG. 3 is a cross-sectional view schematically illustrating a state of a longitudinal cross section taken along line I-I in FIG. 2.

[0030] FIG. 4 is a cross-sectional view schematically illustrating a state of a longitudinal cross section taken along line II-II in FIG. 2.

[0031] FIG. 5 is a cross-sectional view schematically illustrating a state of a longitudinal cross section taken along line III-III in FIG. 2.

[0032] FIG. 6 is a cross-sectional view for explaining an example of an organic layer.

[0033] FIGS. 7A, 7B, and 7C are cross-sectional views for explaining an example of a method of manufacturing the display device.

[0034] FIGS. 8A, 8B, and 8C are cross-sectional views for explaining an example of the method of manufacturing the display device.

[0035] FIGS. 9A, 9B, and 9C are cross-sectional views for explaining an example of the method of manufacturing the display device.

[0036] FIGS. 10A, 10B, and 10C are cross-sectional views for explaining an example of the method of manufacturing the display device.

[0037] FIGS. 11A, 11B, and 11C are cross-sectional views for explaining an example of the method of manufacturing the display device.

[0038] FIGS. 12A, 12B, and 12C are cross-sectional views for explaining an example of the method of manufacturing the display device.

[0039] FIGS. 13A, 13B, and 13C are cross-sectional views for explaining an example of the method of manufacturing the display device.

[0040] FIG. 14 is a plan view for explaining an example of a display device according to a second embodiment.

[0041] FIG. 15 is a plan view for explaining an example of the display device according to the second embodiment.

[0042] FIGS. 16A, 16B, 16C, and 16D are plan views for explaining an example of a display device according to a third embodiment.

[0043] FIGS. 17A, 17B, 17C, and 17D are plan views for explaining an example of the display device according to the third embodiment.

[0044] FIGS. 18A and 18B are diagrams for explaining an example of a display device according to a fourth embodiment.

[0045] FIGS. 19A and 19B are diagrams for explaining an example of the display device according to the fourth embodiment.

[0046] FIGS. 20A and 20B are diagrams for explaining an example of the display device according to the fourth embodiment.

[0047] FIG. 21 is a diagram for explaining an example of the display device according to the fourth embodiment.

[0048] FIGS. 22A, 22B, and 22C are diagrams for explaining an example in a case where the display device includes a wavelength selection unit.

[0049] FIG. 23 is a diagram for explaining an example in a case where the display device includes the wavelength selection unit.

[0050] FIGS. 24A and 24B are diagrams for explaining an example in a case where the display device includes the wavelength selection unit.

[0051] FIG. 25 is a diagram for explaining an example in a case where the display device includes the wavelength selection unit.

[0052] FIG. 26 is a cross-sectional view for explaining an embodiment in a case where the display device includes the wavelength selection unit.

[0053] FIGS. 27A and 27B are diagrams for explaining an application example of the display device.

[0054] FIG. 28 is a diagram for explaining an application example of the display device.

[0055] FIG. 29 is a diagram for explaining an application example of the display device.

[0056] FIG. 30 is a diagram for explaining an application example of the display device.

[0057] FIG. 31 is a diagram for explaining an application example of the display device.

[0058] FIGS. 32A and 32B are diagrams for explaining an application example of the display

device.

[0059] FIG. 33 is a plan view for explaining an example of a display device according to a fifth embodiment.

[0060] FIG. 34 is a cross-sectional view taken along line XXXIV-XXXIV of FIG. 33.

[0061] FIG. 35 is a cross-sectional view taken along line XXXV-XXXV of FIG. 33.

[0062] FIG. 36 is a cross-sectional view taken along line XXXVI-XXXVI of FIG. 33.

[0063] FIG. 37 is a cross-sectional view taken along line XXXVII-XXXVII of FIG. 33.

[0064] FIG. 38 is a plan view of an organic layer.

[0065] FIG. 39 is a plan view of 39 a second electrode.

[0066] FIG. 40 is a plan view for explaining a sub-pixel layout of a display device according to a modification of the fifth embodiment.

[0067] FIGS. 41A and 41B are plan views for explaining a sub-pixel layout of a display device according to a modification of the fifth embodiment.

[0068] FIGS. 42A and 42B are plan views for explaining a sub-pixel layout of a display device according to a modification of the fifth embodiment.

[0069] FIG. 43 is a plan view for explaining a sub-pixel layout of a display device according to a modification of the fifth embodiment.

[0070] FIGS. 44A, 44B, 44C, 44D, and 44E are cross-sectional views for explaining an example of a method of manufacturing the display device.

[0071] FIGS. 45A, 45B, 45C, and 45D are cross-sectional views for explaining an example of the method of manufacturing the display device.

[0072] FIGS. 46A, 46B, 46C, and 46D are cross-sectional views for explaining an example of the method of manufacturing the display device.

[0073] FIGS. 47A, 47B, 47C, and 47D are cross-sectional views for explaining an example of the method of manufacturing the display device.

[0074] FIGS. 48A, 48B, 48C, 48D, and 48E are cross-sectional views for explaining an example of the method of manufacturing the display device.

[0075] FIGS. 49A, 49B, 49C, and 49D are cross-sectional views for explaining an example of the method of manufacturing the display device.

[0076] FIGS. 50A, 50B, 50C, and 50D are cross-sectional views for explaining an example of the method of manufacturing the display device.

[0077] FIGS. 51A, 51B, 51C, and 51D are cross-sectional views for explaining an example of the method of manufacturing the display device.

[0078] FIG. 52 is a plan view for explaining a sub-pixel layout of a display device according to a modification of the fifth embodiment.

[0079] FIG. 53 is a plan view of an organic layer.

[0080] FIG. 54 is a plan view of a second electrode.

MODE FOR CARRYING OUT THE INVENTION

[0081] Hereinafter, an example and the like according to the present disclosure will be described with reference to the drawings. Note that explanation will be made in the following order. In the present specification and the drawings, configurations having substantially the same functional configurations are denoted by the same reference numerals, and redundant descriptions are omitted.

[0082] A light emitting device according to the present disclosure can be used as a display device or the like. Therefore, by taking a case where the light emitting device according to the present disclosure is a display device as an example, description of a light emitting device, a method of manufacturing the light emitting device, and an application example to an electronic device will be continued.

[0083] Note that explanation will be made in the following order. [0084] 1. Display device [0085] 1-1. First embodiment [0086] 1-2. Second embodiment [0087] 1-3. Third embodiment [0088] 1-4. Fourth embodiment [0089] 1-5. Fifth embodiment [0090] 2. Method of manufacturing display

device [0091] 3. Example of case where display device includes wavelength selection unit [0092]

4. Application example

[0093] The following description concerns preferred specific examples of the present disclosure, and the contents of the present disclosure is not limited to these embodiments and the like.

Furthermore, in the following description, directions of front and back, left and right, up and down, and the like are indicated in consideration of convenience of description, but the content of the present disclosure is not limited to these directions. In the examples of FIGS. 1, 2, 3, 4, 5, and the like, it is assumed that a Z-axis direction is an up-down direction (upper side is in a +Z direction, lower side is in a -Z direction.), a Y-axis direction is a front-back direction (front side is in a -Y direction, and back side is in a +Y direction.), and an X-axis direction is a left-right direction (right side is in a +X direction, and left side is in a -X direction.), and the description will be made based on this. This similarity applies to FIGS. 6 to 15, 18 to 21, and 33 to 51. A relative dimensional ratio of the size and thickness of each layer illustrated in each drawing of FIG. 1 and others is illustrated for convenience, and does not limit any actual dimensional ratios. This similarity applies to the drawings of FIGS. 2 to 32 and FIGS. 33 to 51 regarding the definition and the dimensional ratio regarding these directions. A lateral direction (X direction) and a longitudinal direction (Y direction) are examples of a first direction and a second direction orthogonal to each other in a display surface of the display device.

1 Display Device

1-1 First Embodiment

1-1-1 Configuration of Display Device

[0094] A display device **10** according to a first embodiment of the present disclosure has a plurality of emission colors. Furthermore, the display device **10** includes a plurality of pixels, and one pixel is formed by a combination of a plurality of sub-pixels **101** corresponding to a plurality of color types (emission colors). In the display device **10**, the plurality of sub-pixels **101** are two-dimensionally arranged. A connection portion **23** connecting the different sub-pixels **101** is formed in each sub-pixel **101**. The connection portion **23** is formed in a part of an inter-sub-pixel region. The display device **10** includes a second electrode **15** as described later, and the second electrode **15** is formed in the sub-pixel **101** and the connection portion **23**. Then, a portion of the second electrode **15** of the sub-pixel **101** and a portion of the second electrode **15** of the connection portion **23** are continuous.

[0095] Examples of the display device **10** according to the first embodiment of the present disclosure include an organic electroluminescence (EL) display device. In the display device **10** according to the first embodiment, as illustrated in FIGS. 1, 2, 3, 4, and the like, a case where the display device **10** is an organic EL display device **10** (Hereinafter, it is simply referred to as a “display device **10**”.) will be described as an example. FIG. 1 is a plan view illustrating an example of the display device **10**. FIG. 2 is a plan view schematically illustrating a part of an enlarged region XS surrounded by a broken line in FIG. 1. In FIG. 2, for convenience of description, a counter substrate, a protective layer, and the like, which will be described later, are omitted, a second electrode and a multilayer structure are indicated by a solid line, and a section (section line MU) defining a unit region having a sub-pixel is indicated by a two-dot chain line. In a display region **10A**, a plurality of unit regions are arranged in a predetermined layout, and one sub-pixel **101** is formed inside the unit region (inside a hexagonal region partitioned by the section line MU in FIG. 2). FIG. 3 is a cross-sectional view schematically illustrating a state of a cross section taken along line I-I in FIG. 2. FIG. 4 is a cross-sectional view schematically illustrating a state of a cross section taken along line II-II in FIG. 2. FIG. 5 is a cross-sectional view schematically illustrating a state of a cross section taken along line III-III in FIG. 2. Note that, for convenience of description, the counter substrate is not illustrated in FIGS. 3, 4, and 5.

[0096] In the description below, a case where the display device **10** performs display by a top emission method is explained as an example. It is assumed that the top emission method indicates a

method in which light emitting elements **104** are disposed on a side of a light emitting surface DP rather than a side of a substrate **11A**. Accordingly, in the display device **10**, the substrate **11A** is located on a back surface side of the display device **10**, and a direction (+Z direction) from the substrate **11A** toward the light emitting elements **104** described later is a direction toward a front surface side (upper surface side) of the display device **10**. In the display device **10**, light generated from the light emitting elements **104** is directed in the +Z direction, and is emitted to the outside. In the following description, in each layer constituting the display device **10**, a surface on a display surface side in a display region (In FIG. 1, the display region **10A** indicated by a hatched region is illustrated.) of the display device **10** is referred to as a first surface (upper surface), and a surface on a back surface side of the display device **10** is referred to as a second surface (lower surface). Note that this does not prohibit the case where the display device **10** according to the present disclosure is of a bottom emission type. The display device **10** is also applicable to a bottom emission type. By a bottom emission method, light generated from the light emitting elements **104** is directed in the -Z direction, and is emitted to the outside.

[0097] Details of the type of the sub-pixel, the configuration of the sub-pixel, the configuration of the connection portion, and each configuration formed in each sub-pixel will be further described. (Type of Sub-Pixel)

[0098] In the examples of FIGS. 1, 2, 3, 4, 5, and the like, three colors of red, green, and blue are determined as a plurality of color types corresponding to a plurality of emission colors of the display device **10**, and three types of a sub-pixel **101R**, a sub-pixel **101G**, and a sub-pixel **101B** are provided as the sub-pixels **101** corresponding to the plurality of color types. The sub-pixel **101R**, the sub-pixel **101G**, and the sub-pixel **101B** are a red sub-pixel, a green sub-pixel, and a blue sub-pixel, respectively, and display the red color, the green color, and the blue color, respectively. However, the examples of FIGS. 1, 2, 3, 4, 5, and the like are merely examples, and the display device **10** is not limited to a case of including the plurality of sub-pixels corresponding to the three color types. Furthermore, wavelengths of light corresponding to the respective color types of red, green, and blue can be determined as, for example, wavelengths in a range of 610 nm to 650 nm (red wavelength band), a range of 510 nm to 590 nm (green wavelength band), and a range of 440 nm to 480 nm (blue wavelength band), respectively. Note that the number of color types of the sub-pixels **101** is not limited to the three colors illustrated here, and may be two colors, four colors, or the like. Furthermore, the color type of the sub-pixels **101** is not limited to red, green, and blue, and may be yellow, white, or the like.

[0099] Furthermore, the layout of the sub-pixels **101B**, **101R**, and **101G** in the display device **10** is not particularly limited, but in the example of FIG. 2, in the display region **10A**, the sub-pixels **101B**, **101R**, and **101G** constituting one pixel are arranged in a delta shape, and each pixel is two-dimensionally provided. Therefore, in the display device **10** illustrated in the example of FIG. 2, the plurality of sub-pixels **101B**, **101R**, and **101G** corresponding to the plurality of color types is provided in a two-dimensional and delta-shaped layout. The delta-shaped layout indicates a layout in which a triangle is formed by line segments connecting the centers of the plurality of sub-pixels **101** constituting the pixel. Furthermore, in FIG. 2, each of the sub-pixels **101** is defined in a hexagonal shape. Note that FIG. 2 is an example, and as will be described later, the layout and shape of the sub-pixels **101B**, **101R**, and **101G** are not limited in the present disclosure. In FIG. 2, the sub-pixels **101R**, **101B**, and **101G** are denoted by symbols R, G, and B, respectively. Note that, in the display region **10A**, a region between the plurality of sub-pixels **101B**, **101R**, and **101G** (a region outside the sub-pixels **101** in plan view of the display device **10**) is defined as an inter-sub-pixel region M.

[0100] In the description of the present specification, in a case where the types of the sub-pixels **101R**, **101G**, and **101B** are not particularly distinguished, the sub-pixels **101R**, **101G**, and **101B** are collectively referred to as the sub-pixel **101**.

(Drive of Sub-Pixel)

[0101] As illustrated in FIG. 1, the display device **10** generally includes a control circuit (not illustrated), an H driver **105**, and a V driver **106**, and the control circuit **107** controls driving of the H driver **105** and the V driver **106**. The H driver **105** and the V driver **106** control driving of the sub-pixels **101**.

(Configuration of Sub-Pixels)

[0102] In the example of FIGS. 3, 4, 5, and the like, the display device **10** includes a first electrode **13** on an upper side of the drive substrate **11**, and includes an organic layer **14** and a second electrode **15** in order on the first electrode **13**. In this case, the first electrode **13**, the organic layer **14**, and the second electrode **15**, which are sequentially formed on the upper side (+Z direction side) of the drive substrate **11**, form the light emitting elements **104** in the sub-pixel **101**. Note that, in the sub-pixel **101**, a portion of the light emitting elements **104** where the organic layer **14** and the second electrode **15** are layered is referred to as a multilayer structure **22**.

[0103] Next, each configuration of the drive substrate **11** and the like will be described. Note that a configuration that may be common between a configuration of each layer in the sub-pixel **101** and a layer configuration formed in the connection portion **23** described later will be described together.

(Drive Substrate)

[0104] As illustrated in FIGS. 3 to 5 and the like, in the drive substrate **11**, an insulating layer **11B** is provided on the substrate **11A**, and various circuits for driving the plurality of light emitting elements **104** are provided in the insulating layer **11B**. Examples of the various circuits include a drive circuit that controls driving of the light emitting elements **104**, and a power supply circuit that supplies power to the plurality of light emitting elements **104** (none of which is illustrated in the drawings). The various circuits are restricted from being exposed to the outside by the insulating layer **11B**. Furthermore, the drive substrate **11** is provided with a wiring **11C** for connecting the light emitting elements **104**, a circuit provided on the substrate **11A**, and the like to the first electrode **13** and the like. Note that, in FIGS. 3, 4, and 5, the wiring **11C** is illustrated in a state of including a contact plug and the like for convenience of description.

[0105] The substrate **11A** may include, for example, glass or resin having low moisture and oxygen permeability, or may include a semiconductor in which a transistor or the like is easily formed. Specifically, the substrate **11A** may be a glass substrate, a semiconductor substrate, a resin substrate, or the like.

(Insulating Layer)

[0106] The insulating layer **11B** is formed with an organic material or an inorganic material, for example. The organic material contains at least one material of polyimide or acrylic resin, for example. The inorganic material contains at least one material of silicon oxide, silicon nitride, silicon oxynitride, or aluminum oxide, for example.

(Light Emitting Elements)

[0107] The sub-pixel **101** is provided with the plurality of light emitting elements **104** on the first surface of the drive substrate **11**. In the examples of FIGS. 1, 2, 3, 4, 5, and the like, each of the light emitting elements **104** is an organic electroluminescent element (organic EL element). As the plurality of light emitting elements **104**, light emitting elements that set a color corresponding to the color type of the sub-pixel **101** as light emitted (as an emission color) from a light emitting surface (a surface formed by the display region **10A** in FIG. 1) are provided. For example, light emitting elements **104R**, **104G**, and **104B** are formed in the sub-pixels **101R**, **101G**, and **101B**, respectively. Furthermore, the plurality of light emitting elements **104** is arranged corresponding to the arrangement of the sub-pixels **101** of the respective color types. In the examples of FIGS. 2, 3, 4, and 5, the plurality of light emitting elements **104** is two-dimensionally arranged in a delta-shaped arrangement pattern. Note that, in the present specification, in a case where the types such as the light emitting elements **104R**, **104G**, and **104B** are not particularly distinguished, the term light emitting element **104** is used.

[0108] Each of the light emitting elements **104** has a structure in which the first electrode **13**, the

organic layer **14**, and the second electrode **15** are layered in this order. The first electrode **13**, the organic layer **14**, and the second electrode **15** are layered in this order from a side of the drive substrate **11** in a direction (+Z direction) from the second surface toward the first surface.

(First Electrode)

[0109] A plurality of the first electrodes **13** is provided on a side of the first surface of the drive substrate **11**. In the examples of FIGS. **3**, **4**, **5**, and the like, each of the first electrodes **13** is an anode electrode.

[0110] The first electrodes **13** each include at least one of a metal layer or a metal oxide layer. The first electrodes **13** may each include a single-layer film of a metal layer or a metal oxide layer, or a multilayer film of a metal layer and a metal oxide layer.

[0111] The metal layer contains at least one metal element selected from the group consisting of chromium (Cr), gold (Au), platinum (Pt), nickel (Ni), copper (Cu), molybdenum (Mo), titanium (Ti), tantalum (Ta), aluminum (Al), magnesium (Mg), iron (Fe), tungsten (W), and silver (Ag), for example. The metal layer may contain the at least one metal element described above as a constituent element of an alloy. Specific examples of the alloy include an aluminum alloy and a silver alloy. Specific examples of the aluminum alloy include AlNd and AlCu, for example.

[0112] The metal oxide layer contains at least one of a mixture of indium oxide and tin oxide (ITO), a mixture of indium oxide and zinc oxide (IZO), or titanium oxide (TiO), for example.

[0113] In the examples of FIGS. **2**, **3**, **4**, **5**, and the like, the first electrode **13** is formed at least in each of the sub-pixels **101**, and is electrically separated for each of the sub-pixels **101**. That is, a plurality of the first electrodes **13** is provided on the side of the first surface of the drive substrate **11**, and is provided for each of the sub-pixels **101**. In the examples of FIGS. **2**, **3**, **4**, and **5**, the first electrode **13** is formed slightly outside the sub-pixels **101**, but is kept in a state of being electrically separated for each of the sub-pixels **101**.

[0114] Furthermore, a layer having insulating properties is preferably formed between the first electrodes **13** adjacent to each other. In the examples of FIGS. **2**, **3**, **4**, **5**, and the like, an insulating layer **12** is formed between the adjacent first electrodes **13**. The insulating layer **12** may be a layer including an inorganic insulating material such as SiO₂, SiN, or SiON formed by a CVD method, a layer including Al₂O₃ formed by an ALD method, or a layer including an organic insulating material such as polyimide. The insulating layer **12** may be a layer having a single layer structure or a layer having a multilayer structure. The insulating layer **12** may be a layer formed with the same material as the insulating layer **11B** or a layer formed with a material different from the insulating layer **11B**. In a case where the insulating layer **12** is the same as the insulating layer **11B**, the insulating layer **12** may be integrated with the insulating layer **11B**. In the examples of FIGS. **2**, **3**, **4**, **5**, and the like, the insulating layer **12** electrically separates each first electrode **13** for each light emitting element **104** (for each sub-pixel **101**). Furthermore, as illustrated in FIGS. **3**, **4**, and the like, an opening **12A** is formed in the insulating layer **12** on the side of the first surface, the first surface of the first electrode **13** (a surface facing the second electrode **15**) is exposed from the opening **12A** of the insulating layer **12**, and a portion of the first electrode **13** exposed from the opening **12A** faces the organic layer **14** to be described later without interposing the insulating layer **11B**. Note that the insulating layer **11B** may be formed not only between the adjacent first electrodes **13**, but also onto the edges of the first electrodes **13**. The edge of each of the first electrode **13** is defined by a portion from an outer peripheral edge of the first electrode **13** to a predetermined position closer to a center side of the first electrode **13**. Also in this case, the insulating layer **11B** includes the opening **12A**, and the first surface of the first electrode **13** is exposed from the opening **12A**.

(Organic Layer)

[0115] The organic layer **14** is provided on an upper side of the first electrode **13**. The organic layer **14** is formed at least in each sub-pixel **101**. In the sub-pixel **101**, the organic layer **14** is provided between the first electrode and the second electrode **15**. The organic layer **14** is an organic

compound layer, and is provided according to the color type of the sub-pixel **101**. For example, organic layers **14R**, **14G**, and **14B** are formed in accordance with the sub-pixels **101R**, **101G**, and **101B**, respectively. The organic layer **14R** is the organic layer **14** having red as an emission color. The organic layer **14G** is the organic layer **14** having green as an emission color. The organic layer **14B** is the organic layer **14** having blue as an emission color. In the present specification, in a case where the types such as the organic layers **14R**, **14G**, and **14B** are not particularly distinguished, the term organic layer **14** is used.

[0116] As illustrated in FIG. 6, the organic layer **14** illustrated in FIGS. 3, 4, 5, and the like has a structure in which a light emitting layer **142** and a plurality of functional layers **25** as other layers excluding the light emitting layer **142** are layered. FIG. 6 is a diagram illustrating an example of a layer configuration of the light emitting element **104**. Note that FIG. 6 illustrates the light emitting element **104B** as an example. In the example illustrated in FIG. 6, the organic layer **14** has a configuration in which a hole injection layer **140**, a hole transport layer **141**, the light emitting layer **142**, and an electron transport layer **143** are layered in this order from the first electrode **13** toward the second electrode **15** (from a side closer to the first electrode **13**). As illustrated in FIG. 6, an electron injection layer **144** may be provided between the electron transport layer **143** and the second electrode **15**. In this case, in the organic layer **14**, the functional layers **25** excluding the light emitting layer **142** are the hole injection layer **140**, the hole transport layer **141**, the electron transport layer **143**, and the electron injection layer **144**. Note that, in the present specification, for convenience of description, the functional layers **25** such as the hole injection layer **140** and the hole transport layer **141** formed between the light emitting layer **142** and the first electrode **13** in the organic layer **14** are collectively referred to as a first layer **125A**, and the functional layers **25** such as the hole injection layer **140** and the hole transport layer **141** formed between the light emitting layer **142** and the second electrode **15** in the organic layer **14** are collectively referred to as a second layer **125B**.

[0117] Note that, in FIG. 4, for convenience of description, the first layer **125A** and the second layer **125B** are described, and description of the hole injection layer **140**, the hole transport layer **141**, the electron transport layer **143**, and the electron injection layer **144** is omitted. Furthermore, in FIGS. 3 and 5, for convenience of explanation, a structure in which the plurality of functional layers **25** excluding the light emitting layer **142** from the organic layer **14** is layered is displayed as a layer **126** without distinguishing the first layer **125A** and the second layer **125B**. The layer **126** indicates a layer structure of a portion obtained by removing the light emitting layer **142** from the organic layer **14**.

[0118] The hole injection layer **140** is a buffer layer for enhancing efficiency of hole injection into the light emitting layer **142** and reducing leakage. Examples of the material of the hole injection layer **140** can include hexaazatriphenylene (HAT). The hole transport layer **141** is for enhancing efficiency of hole transport to the light emitting layer **142**. Examples of the material of the hole transport layer **141** can include N,N'-di(1-naphthyl)-N,N'-diphenyl-1,1'-biphenyl-4,4'-diamine (α -NPD).

[0119] The electron transport layer **143** is for enhancing efficiency of electron transport to the light emitting layer **142**. Examples of the material of the electron transport layer **143** can include aluminum quinolinol, bathophenanthroline, and the like.

[0120] Note that the electron injection layer **144** is for enhancing electron injection efficiency. Examples of the material of the electron injection layer **144** can include a simple substance of an alkali metal or an alkaline earth metal, such as lithium or lithium fluoride, and a compound including those simple substances.

[0121] The light emitting layers **142** generate light by recombination of electrons and holes when an electric field is applied. The light emitting layers **142** are organic compound layers containing an organic light emitting material. As light emitting layers **142R**, **142G**, and **142B** of the organic layers **14R**, **14G**, and **14B**, a layer containing an organic light emitting material corresponding to

each emission color is suitably used. For example, in the light emitting layer **142R** of the organic layer **14R**, a layer containing a red light emitting material (red light emitting layer) can be suitably used. Specifically, as the red light emitting material, a material obtained by mixing 4,4-bis(2,2-diphenylvinyl) biphenyl (DPVBi) with 30 weight % of 2,6-bis[(4'-methoxydiphenylamino)styryl]-1, 5-dicyanonaphthalene (BSN) can be used. In the light emitting layer **142G** of the organic layer **14G** and the light emitting layer **142B** of the organic layer **14B**, a layer containing a green light emitting material (green light emitting layer) and a layer containing a blue light emitting material (blue light emitting layer) can be suitably used, respectively. The green light emitting material is not particularly limited, and an organic light emitting material capable of emitting green light may be used. Examples of the green light emitting material include a mixture of DPVBi and coumarin 6. As the blue light emitting material, similarly to the red light emitting material, the green light emitting material, or the like, an organic light emitting material capable of emitting blue light may be used. Examples of the blue light emitting material include a mixture of DPVBi and 4,4-bis(2-(4-(N,N-diphenylamino) phenyl) vinyl) biphenyl (DPAVBi).

[0122] Note that, in the example of FIG. 5, the organic layer **14** includes the light emitting layer **142** as a single layer, but may include a plurality of the light emitting layers **142**. In a case where the plurality of light emitting layers **142** are provided, a layer other than the light emitting layers may be provided as the functional layer **25** between the light emitting layers **142**.

(Correlation Between Functional Layers of Sub-Pixels Corresponding to Respective Color Types)

[0123] For each of the functional layers **25** defined as a layer of the organic layer **14** excluding the light emitting layer **142**, a common layer may be adopted regardless of the color type of the sub-pixel **101**, a partially different layer may be adopted, or a completely different layer may be adopted.

[0124] For example, any of the hole injection layer **140**, the hole transport layer **141**, the electron transport layer **143**, and the electron injection layer **144** may be a layer common to the sub-pixel **101R**, the sub-pixel **101B**, and the sub-pixel **101G**. The hole injection layer **140** and the hole transport layer **141** may be different for each of the sub-pixel **101R**, the sub-pixel **101B**, and the sub-pixel **101G**, and the electron transport layer **143** and the electron injection layer **144** may be layers common to the sub-pixel **101R**, the sub-pixel **101B**, and the sub-pixel **101G**. Furthermore, any of the hole injection layer **140**, the hole transport layer **141**, the electron transport layer **143**, and the electron injection layer **144** may be different layers for each of the sub-pixel **101R**, the sub-pixel **101B**, and the sub-pixel **101G**. From the viewpoint of simplifying the manufacturing process, it is preferable that at least some functional layers **25** are common to the multilayer structure **22** of the plurality of sub-pixels **101** regardless of the color type of the sub-pixel **101**.

[0125] In the examples of FIGS. 3 and 4, the hole transport layer **141** of the functional layers **25** may be different in thickness from each of the sub-pixel **101R**, the sub-pixel **101B**, and the sub-pixel **101G**. Furthermore, in the examples of FIGS. 3 and 4, any of the hole injection layer **140**, the electron transport layer **143**, and the electron injection layer **144** may be a layer common to the sub-pixel **101R**, the sub-pixel **101B**, and the sub-pixel **101G**.

(Layer Formed at Connection Portion in Organic Layer)

[0126] In the examples of FIGS. 3 and 5, at least a part of the functional layers **25** defined as a layer excluding the light emitting layer in the organic layer **14** is formed in the sub-pixel **101** and the connection portion **23**. Each of the functional layers **25** formed in the connection portion **23** may be a layer common to the functional layer **25** formed in one sub-pixel **101**, may be a completely different layer, or may be a partially different layer.

[0127] In the example of FIG. 3, as an example of the connection portion **23** connecting the sub-pixel **101R** and the sub-pixel **101B**, as also illustrated in FIG. 11C, a first portion **122** of the connection portion **23** may be common to a combination (first layer **125A**, second layer **125B**, and layer **126**) of the functional layers **25** formed in the sub-pixel **101R**, and a second portion **123** may be common to a combination (first layer **125A**, second layer **125B**, and layer **126**) of the functional

layers **25** formed in the sub-pixel **101R**.

[0128] In the examples of FIGS. **3** and **5**, the light emitting layer **142** of the organic layer **14** extends from one sub-pixel **101** to the connection portion **23** connected to the other sub-pixel **101**. The light emitting layer **142** may extend to the connection portion **23** for each of the connection portions **23** connected to one sub-pixel **101**. In the example of FIG. **3**, the light emitting layer **142B** provided in the sub-pixel **101B** extends to both the connection portion **23** connected to a side of the sub-pixel **101R** and the connection portion **23** connected to a side of the sub-pixel **101G**. Note that, in a case where the light emitting layer **142** extends to the connection portion **23**, it is preferable that the light emitting layer **142** only extends to a part of the connection portion **23**. In this case, the light emitting layer **142** is included in a part of the connection portion **23**. Note that the example of FIG. **3** is an example, and it is not prohibited that the light emitting layer **142** of the organic layer **14** does not extend from the one sub-pixel **101** to the connection portion **23** connected to the other sub-pixel **101**.

[0129] Furthermore, in a case where the light emitting layer **142** extends to the connection portions **23** for the plurality of connection portions **23**, the light emitting layer **142** extending to some of the connection portions **23** may be different from the light emitting layer extending to the other connection portions **23**. Furthermore, a plurality of types of the light emitting layers **142** may extend to one connection portion **23**. Furthermore, as illustrated in FIG. **3**, in a case where the plurality of light emitting layers **142** extending to the connection portion **23** is formed, the combination of the light emitting layers **142** may be different depending on the connection portion **23**. For example, in a case where the light emitting layer **142** extending to some of the connection portions **23** is a layer having a red color as an emission color, the light emitting layer **142** extending to the other connection portions **23** may be a layer having a green color as an emission color. For example, as for the connection portion **23** illustrated in FIG. **3**, one in which the light emitting layer **142B** and the light emitting layer **142R** extend to the connection portion **23** and one in which the light emitting layer **142B** and the light emitting layer **142G** extend to the connection portion **23** are illustrated. In FIG. **5**, the light emitting layer **142G** and the light emitting layer **142R** extend to the connection portion **23**. As described above, in a case where the light emitting layers **142** extend to the connection portion **23**, the combination of the light emitting layers **142** may be different depending on the arrangement of the connection portion **23**.

[0130] In the connection portion **23**, the light emitting layers **142** may overlap. For example, in the example of FIG. **3**, in the connection portion **23** connecting the sub-pixel **101R** and the sub-pixel **101B**, the light emitting layer **142** of the organic layer **14R** and the light emitting layer **142** of the organic layer **14B** extend. In the connection portion **23** connecting the sub-pixel **101B** and the sub-pixel **101G**, the light emitting layer **142B** of the organic layer **14B** and the light emitting layer **142G** of the organic layer **14G** extend. Moreover, at the connection portion **23**, the light emitting layer **142G** and the light emitting layer **142B** overlap each other.

(Second Electrode)

[0131] The second electrode **15** is provided on an upper side (first surface side) of the organic layer **14**. A portion of the second electrode **15** corresponding to the sub-pixel **101** (a portion corresponding to the light emitting element **104**) is provided so as to face the first electrode **13**. In the examples of FIGS. **2**, **3**, **4**, **5**, and the like, the second electrode **15** is provided as an electrode common to the plurality of sub-pixels **101** corresponding to the plurality of emission colors. The second electrode **15** is formed at least in the plurality of sub-pixels **101** and the connection portion **23**. The second electrode **15** is formed in common and continuously with at least some of the sub-pixels **101** and the connection portions **23** connected to the sub-pixels **101**. This can be realized, for example, by patterning the second electrode **15** in a layout corresponding to a combined portion of the sub-pixel **101** and the connection portion **23** using photolithography and etching as described in a manufacturing method to be described later.

[0132] The second electrode **15** is a cathode electrode. The second electrode **15** is preferably a

transparent electrode having transparency to light generated in the organic layer **14**. The transparent electrode herein may be a transparent electrode formed with a transparent conductive layer, or a transparent electrode formed with a multilayer structure including a transparent conductive layer and a semi-transmissive reflective layer.

[0133] As the transparent conductive layer, a transparent conductive material having good optical transparency and a small work function is preferably used. The transparent conductive layer can be formed with a metal oxide, for example. Specifically, examples of the material of the transparent conductive layer can include a material containing at least one of a mixture of indium oxide and tin oxide (ITO), a mixture of indium oxide and zinc oxide (IZO), or zinc oxide (ZnO).

[0134] The semi-transmissive reflective layer can be formed with a metal layer, for example. Specifically, examples of the material of the semi-transmissive reflective layer can include a material containing at least one metal element selected from the group consisting of magnesium (Mg), aluminum (Al), silver (Ag), gold (Au), and copper (Cu). The metal layer may contain the at least one metal element described above as a constituent element of an alloy. Specific examples of the alloy include an MgAg alloy, an AgPdCu alloy, and the like.

(First Protective Layer)

[0135] In the examples illustrated in FIGS. **3**, **4**, **5**, and the like, a first protective layer **16** is preferably formed as a protective layer so as to cover the first surface of the light emitting element **104** (exposed surface of the second electrode **15**). The first protective layer **16** makes it difficult for the first surface of the light emitting element **104** to come into contact with the outside air, and suppresses moisture ingress into the light emitting element **104** from the external environment. The first protective layer **16** has translucency with respect to light emitted from the light emitting element **104**.

[0136] The first protective layer **16** is formed with an insulating material. As the insulating material, thermosetting resin or the like can be used, for example. As the insulating material forming the first protective layer **16**, an organic insulating material such as polyimide may be used. In addition, as the insulating material, an inorganic insulating material such as SiO.sub.2, SiON, AlO, or TiO may be used. In this case, as the first protective layer **16**, a CVD film containing SiO.sub.2, SiON, or the like, an ALD film containing AlO, TiO, SiO.sub.2, or the like, or the like can be exemplified. Note that the CVD film indicates a film formed using chemical vapor deposition. The ALD film indicates a film formed using atomic layer deposition.

(Second Protective Layer)

[0137] A second protective layer **17** is preferably formed as a protective layer so as to cover a side of the first surface of the first protective layer **16** and between the light emitting elements **104** adjacent to each other. In the examples of FIGS. **3**, **4**, **5**, and the like, the second protective layer **17** is formed on one surface so as to cover a formation region of the sub-pixel **101** and an inter-sub-pixel region. The second protective layer **17** may be formed of an insulating material similar to that of the first protective layer **16**. Similarly to the first protective layer **16**, the second protective layer **17** makes it difficult for the first surface of the light emitting element **104** to come into contact with the outside air, and suppresses moisture ingress into the light emitting element **104** from the external environment. The second protective layer **17** has translucency with respect to light emitted from the light emitting element **104**.

[0138] Note that, in FIGS. **3**, **4**, **5**, and the like, the first protective layer **16** and the second protective layer **17** are illustrated separately, but the first protective layer **16** and the second protective layer **17** may form one layer.

(Side Wall of Multilayer Structure)

[0139] As illustrated in FIG. **4**, the multilayer structure **22** in the light emitting element **104** formed in each sub-pixel **101** includes a side wall **24**. The side wall **24** includes a side end surface **241** of the organic layer **14** and a side end surface **242** of the second electrode **15**. Furthermore, in each sub-pixel **101**, the side wall **24** is covered with the second protective layer **17**.

[0140] In at least a part of the sub-pixel **101**, the side end surface **241** of the organic layer **14** and the side end surface **242** of the second electrode **15** are preferably aligned at a position of a boundary between the organic layer **14** and the second electrode **15**. In the example of FIG. **4**, in at least a part of the sub-pixel **101**, the side end surface **241** of the organic layer **14** and the side end surface **242** of the second electrode **15** form a substantially continuous surface at the position of the boundary between the organic layer **14** and the second electrode **15**.

[0141] Furthermore, in the side end surface **241** of the organic layer **14** of the side wall **24** of the multilayer structure **22**, a side end surface of the light emitting layer **142** and side end surfaces of the plurality of functional layers **25** are preferably aligned. In the example of FIG. **4**, in at least a part of the sub-pixel **101**, the side wall **24** of the multilayer structure **22** forms a surface that is substantially continuous in the up-down direction, and the side end surface of the light emitting layer **142** and the side end surfaces of the plurality of functional layers **25** form a substantially continuous surface.

[0142] As described above, the first protective layer **16** is formed on the upper side of the multilayer structure **22** (the upper side of the second electrode **15**). In a case where the first protective layer **16** and the second protective layer **17** formed in each sub-pixel **101** are distinguished, the first protective layer **16** includes a side end surface **243**. In this case, in at least a part of the sub-pixel **101**, the side end surface **243** of the first protective layer **16** and the side end surface **242** of the second electrode **15** are preferably aligned at least at the position of the boundary between the first protective layer **16** (protective layer) and the second electrode **15**. In the example of FIG. **4**, the side wall **24** of the multilayer structure **22** and the side end surface of the first protective layer **16** form a substantially continuous surface, and the side end surface of the first protective layer **16** and the side end surface of the second electrode **15** form a substantially continuous surface.

(Connection Portion)

[0143] The display device **10** includes the connection portion **23** as described above, and the connection portion **23** is defined as a portion that connects a plurality of the multilayer structures **22** formed in different sub-pixels **101** and is two-dimensionally arranged at positions between the plurality of different multilayer structures **22** as illustrated in FIGS. **2**, **3**, **5**, and the like. In the example of FIG. **2**, the connection portion **23** is arranged in an inter-sub-pixel region (an outer region of the multilayer structures **22** in plan view of the display device **10**). Furthermore, in this example, the layer (for example, the hole injection layer **140**) disposed closest to a side of the second surface among the functional layers **25** constituting the organic layer **14** formed in the connection portion **23** is in contact with the insulating layer **12**. Note that the plan view of the display device **10** indicates a case where the Z-axis direction is viewed as a line-of-sight direction.

(Configuration of Connection Portion)

[0144] The connection portion **23** includes at least the second electrode **15**. In the examples of FIGS. **3** and **5**, as described above, the connection portion **23** further includes the layer **126** (functional layer **25**) excluding the light emitting layer **142** in the organic layer **14**. Furthermore, in this example, a part of the light emitting layer **142** extends to the connection portion **23**, and a part of the connection portion **23** includes a part of the light emitting layer **142**. However, the structure of the connection portion **23** illustrated in FIGS. **3** and **5** is merely an example, and does not limit the connection portion **23** of the display device **10**.

(Layout of Connection Portion)

[0145] The layout and shape of the connection portion **23** are not particularly limited, but in the example of FIG. **2**, in at least some of the sub-pixels **101**, the plurality of connection portions **23** are connected to different positions on the outer peripheral surface of the multilayer structure **22**. In the example of FIG. **2**, in the sub-pixel **101**, the connection portions **23** are connected to six different positions.

[0146] As illustrated in the example of FIG. **2**, the connection portion **23** may be formed in a layout

connecting the multilayer structures **22** provided in the sub-pixels **101** having different emission colors. For example, in FIG. **2**, the connection portion **23** is formed in a rectangular shape, and has one end side connected to the multilayer structure **22** of the sub-pixel **101B** and the other end side connected to the multilayer structure **22** provided in the sub-pixel **101R** or the sub-pixel **101G**. However, this does not deny that the connection portion **23** is formed in a layout connecting the multilayer structures **22** provided in the sub-pixels **101** having the same emission color. Furthermore, the connection portion **23** may be formed in a layout in which a layout connecting the multilayer structures **22** provided in the sub-pixels **101** having the same emission color and a layout connecting the multilayer structures **22** provided in the sub-pixels **101** having different emission colors are combined.

[0147] In the examples of FIGS. **2**, **3**, and **5**, the connection portion **23** is formed in a layout connecting the multilayer structures **22** of the two different sub-pixels **101**, but is not limited thereto. The connection portion **23** may be formed in a layout connecting the multilayer structures **22** of three or more different sub-pixels **101** as described later.

[0148] A shape of the connection portion **23** is not particularly limited as long as the resistance of the second electrode **15** is not significantly affected. In the example of FIG. **2**, in plan view of the display device **10**, the shape of the connection portion **23** is generally rectangular and linear, but is not limited thereto, and may be a cross shape, a comb shape, a triangular shape, a circular shape, or the like, or a shape extending in a non-linear shape. Furthermore, a thickness of the connection portion **23** is not particularly limited as long as the resistance of the second electrode **15** is not significantly affected. For example, in a case where the plurality of connection portions **23** are connected to the multilayer structure **22**, the thickness of some of the connection portions **23** may be different from a thickness of the other connection portions **23**, or the thicknesses of the connection portions **23** may be different from each other.

(Low Refractive Index Layer)

[0149] A low refractive index layer **18** is preferably provided on the first surface of the second protective layer **17**. In the examples of FIGS. **2**, **3**, **4**, and **5**, one surface of the low refractive index layer **18** is formed on a side of the first surface of the second protective layer **17**.

[0150] The low refractive index layer **18** is preferably a layer having a smaller refractive index than the protective layer (the first protective layer **16** or the second protective layer **17**). The refractive index of the low refractive index layer **18** is preferably about less than 1.7. Examples of the material forming the low refractive index layer **18** include an ultraviolet curable resin and a thermosetting resin.

[0151] Since the low refractive index layer **18** is provided, interface reflection (interface reflection between the protective layer and the low refractive index layer **18**) of light emitted laterally from the light emitting element **104** can be increased by a difference in refractive index between the protective layer and the low refractive index layer **18**. As a result, light leakage to the adjacent sub-pixel **101** can be suppressed, and light extracted to the front can be increased. Note that light emitted upward from the light emitting element **104** is incident on the interface between the protective layer and the low refractive index layer **18** at a perpendicular or shallow angle, and thus is hardly affected by the difference in refractive index between the protective layer and the low refractive index layer **18**. Therefore, extraction of light emitted upward from the light emitting element **104** hardly decreases due to the difference in refractive index between the protective layer and the low refractive index layer **18**.

(Counter Substrate)

[0152] On the side of the first surface of the low refractive index layer **18**, a counter substrate may be provided (not illustrated). As the material of the counter substrate, the material of the substrate **11A** of the drive substrate **11** or the like can be used. For example, a glass substrate can be used as the counter substrate. The material of the glass substrate is not limited to any particular material, as long as the glass substrate is formed with a material that transmits light emitted from the organic

layer **14**. Examples of the material of the glass substrate include various glass substrates such as high strain point glass, soda glass, borosilicate glass, and lead glass, and quartz substrates.

1-1-2 Operation and Effect

[0153] In a case where the display device includes a plurality of sub-pixels having a plurality of emission colors and different emission colors, a portion (overlapping portion) in which the second electrodes formed in the sub-pixels corresponding to the different emission colors intersect each other is generated, and a step is sometimes formed in the overlapping portion. In such a display device, the respective second electrodes are formed using a manufacturing process in which the second electrodes constituting the sub-pixels corresponding to different emission colors are separated from each other in the overlapping portion.

[0154] In the display device **10** according to the present disclosure, the second electrodes **15** formed in the multilayer structures **22** of the plurality of different sub-pixels **101** are connected at the connection portion **23**, and the second electrodes **15** are common (continuous) to the connection portion **23** and the multilayer structures **22**. In this case, the plurality of different sub-pixels **101** may be the sub-pixels **101** corresponding to different emission colors. Therefore, in the display device **10**, it is not necessary to form the second electrodes **15** provided in the sub-pixels **101** corresponding to different emission colors for each emission color, and the second electrodes **15** provided in the sub-pixels **101** corresponding to the plurality of emission colors can be collectively formed, so that the manufacturing process can be simplified.

[0155] Furthermore, in the display device **10**, since the second electrodes **15** constituting the sub-pixels corresponding to different emission colors can be collectively formed, the overlapping portion of the second electrodes can be omitted. As a result, in the display device **10**, it is possible to suppress generation of a step (unevenness) accompanying the overlapping portion, and it is possible to suppress an increase in resistance and deterioration of optical characteristics of the second electrode in a case where the step is generated. Then, in the display device **10**, the increase in resistance of the second electrode can be suppressed, so that the occurrence of luminance unevenness can be suppressed.

[0156] Furthermore, in the display device **10**, since the connection portion **23** is formed in a pattern so as to have a predetermined layout, a leakage current can be made smaller than in a case where the connection portion is formed in the entire inter-sub-pixel region.

1-2 Second Embodiment

[0157] In the display device **10** of the first embodiment, the layout of the connection portion **23** connected to the sub-pixels **101R**, **101G**, and **101B** is not limited to the example of FIG. 2, and may be a different layout from the example of FIG. 2 as illustrated in FIGS. 14 and 15. An embodiment in which the layout of the connection portion **23** is different from that in the example of FIG. 2 is referred to as a second embodiment. FIGS. 14 and 15 are diagrams illustrating an example of a layout of a connection portion **23** in a display device **10** according to the second embodiment.

[0158] In the display device **10** according to the second embodiment, for example, as illustrated in FIG. 14, the layout of the connection portion **23** may be a layout in which the connection portion **23** is connected to four different locations of one sub-pixel **101**. As illustrated in FIG. 15, the layout of the connection portion **23** may be a layout connected to two different locations of one sub-pixel **101**.

[0159] Furthermore, the layout of the connection portion **23** may be a layout connected to seven or more different locations of one sub-pixel **101**. In the display device **10**, in a case where one sub-pixel **101** is connected to a plurality of other sub-pixels **101** by the connection portion **23**, resistance by the connection portion **23** can be suppressed.

1-3 Third Embodiment

[0160] In the display device **10** of the first embodiment, the layout (layout of the multilayer structures **22**) of the sub-pixels **101R**, **101G**, and **101B** is not limited to the example of FIG. 1

(delta type layout), and may be a layout different from the example of FIG. 1 as illustrated in FIGS. 16 and 17. An embodiment in which the layout of sub-pixels 101 is different from that in the example of FIG. 1 is referred to as a third embodiment. FIGS. 16 and 17 are diagrams illustrating an example of a layout of multilayer structures 22 of the sub-pixels 101 in a display device according to the third embodiment.

(Layout of Sub-Pixels)

[0161] In the display device 10 according to the third embodiment, for example, as illustrated in FIGS. 16A, 16C, 17A, 17B, and 17C, the sub-pixels 101 may be arranged in a stripe type layout. The sub-pixels 101 may be arranged in a square type layout as illustrated in FIGS. 16B, 16D, and 16D.

[0162] The stripe type layout indicates a layout in which a plurality of sub-pixels 101 constituting one pixel are arranged side by side. The square type layout indicates a layout in which centers of the plurality of sub-pixels 101 constituting one pixel are arranged so as to be substantially at a rectangular vertex position (vertex positions of squares in examples of FIGS. 16B and 16D). The similarity applies to FIG. 17.

(Connection Portion)

[0163] Also, in the display device 10 according to the third embodiment, a plurality of connection portions 23 are connected to the sub-pixels 101 at different positions. In FIG. 16A, the different sub-pixels 101 are connected in the lateral direction (X direction) by the connection portions 23. In FIG. 16B, the different sub-pixels 101 are connected in the lateral direction and the longitudinal direction (the X direction and the Y direction) by the connection portions 23. In FIGS. 16A, 16B, and 16C, the sub-pixels 101 corresponding to different emission colors are connected by the connection portions 23. In the example of FIG. 16D, a sub-pixel 101B having two pixels and sub-pixels 101R and 101G are arranged in a square shape. Then, the connection portions 23 are formed in a cross shape so as to connect the sub-pixel 101B and the sub-pixels 101R and 101G. In FIG. 16D, by the connection portion 23, not only the sub-pixels 101 (sub-pixels 101R, 101B, and 101G) corresponding to different emission colors are connected, but also the plurality of sub-pixels 101 (sub-pixel 101B, 101B) corresponding to the same emission color is connected.

[0164] The layout of the connection portions 23 in FIG. 17A, is a combination of the layouts of the connection portions 23 as illustrated in FIGS. 16A and 16C. The layout of the connection portions 23 in FIG. 17D is a combination of the layouts of the connection portions 23 as illustrated in FIGS. 16B and 16D.

[0165] In FIG. 17B, the connection portions 23 connecting the sub-pixels 101 adjacent to each other in the lateral direction (X direction) connect the sub-pixels 101 corresponding to different emission colors, but the connection portions 23 connecting the multilayer structures 22 adjacent to each other in the longitudinal direction (Y direction) connect the sub-pixels 101 corresponding to the same emission color.

[0166] In FIG. 17C, one connection portion 23 connects three or more sub-pixels 101.

1-4 Fourth Embodiment

[0167] A display device 10 according to a fourth embodiment will be described. In the display device 10 according to the fourth embodiment, a resonator structure is further formed in at least a part of the plurality of sub-pixels 101 in the first embodiment. The second embodiment or the third embodiment may be applied to the display device 10 according to the fourth embodiment. In the display device 10 according to the fourth embodiment, other configurations except for the resonator structure may be similar to those of the first to third embodiments, and thus description of other points except for the resonator structure is omitted.

(Resonator Structure)

[0168] A resonator structure is formed in the display device 10. The resonator structure is a cavity structure and causes resonance of light emitted from an organic layer 14. In the display device 10, the resonator structure is formed in each of light emitting elements 104 (light emitting elements

104R, **104B**, and **104G**), and the resonator structure includes a first electrode **13**, the organic layer **14**, and a second electrode **15**. Causing resonance of light emitted from the organic layer **14** means causing resonance of light of a specific wavelength included in the emitted light.

[0169] In the resonator structure, a component reflected and resonated between predetermined layers such as between the first electrode **13** and the second electrode **15** among the light emitted from the organic layer **14** is emphasized, and the light emphasized from a side of the light emitting surface (first surface side) is emitted toward the outside.

[0170] The organic layer **14** generally uses light corresponding to the color type of the sub-pixel **101** as emitted light, and the resonator structure causes resonance of light of a specific wavelength included in the emitted light from the organic layer **14**. In this case, light of a predetermined wavelength in the emitted light from the organic layer **14** is emphasized. Then, the light of the predetermined wavelength being emphasized, light is emitted toward the outside from the side of the second electrode **15** (which is the side of the light emitting surface) of the light emitting element **104**. Note that the light of the predetermined wavelength is light corresponding to a predetermined color type, and indicates light corresponding to a color type determined in accordance with the sub-pixel **101**. The display device **10** includes light emitting elements **104R**, **104G**, and **104B** corresponding to sub-pixels **101R**, **101G**, and **101B**. Furthermore, a resonator structure is formed for each of the light emitting elements **104R**, **104G**, and **104B**. In the resonator structure in the sub-pixel **101R**, red light of the emitted light from the organic layer **14** resonates. Light is emitted toward the outside from the second electrode **15** of the light emitting element **104R**, with the red light being emphasized. Accordingly, the red light having excellent color purity can be emitted from the sub-pixel **101R**. In the resonator structures in the sub-pixels **101G** and **101B**, green light and blue light of the emitted light from the organic layer **14** resonate, respectively. In the sub-pixels **101G** and **101B**, light is emitted toward the outside from the second electrodes **15** of the light emitting elements **104G** and **104B**, with the green light and the blue light being emphasized. Accordingly, the green light and the blue light having excellent color purity can be emitted from the sub-pixels **101G** and **101B**, respectively.

[0171] As the resonator structures are formed in the display device **10** in this manner, the color purity of the sub-pixels **101** can be enhanced.

[0172] First to seventh examples will be sequentially described below as example cases where the display device **10** includes resonator structures, and the explanation will be continued.

(Resonator Structure: First Example)

[0173] FIG. **18A** is a schematic cross-sectional view for explaining a first example in a case where the display device **10** has a resonator structure.

[0174] In the first example, a thickness of the first electrode **13** and a thickness of the second electrode **15** are uniform among the sub-pixels **101R**, **101G**, and **101B**.

[0175] In each of the sub-pixels **101R**, **101G**, and **101B** (light emitting elements **104R**, **104G**, and **104B**), an optical adjustment layer **31** is provided on a lower side (second surface side) of the first electrode **13**, a reflector **30** is further disposed on the second surface side of the optical adjustment layer **31**, and the optical adjustment layer **31** is formed between the reflector **30** and the first electrode **13**. A resonator structure that resonates light generated by the organic layer **14** (organic layers **14R**, **14G**, and **14B**) is formed between the reflector **30** and the second electrode **15**.

[0176] A thickness of the reflector **30** is uniform in the sub-pixels **101R**, **101G**, and **101B**.

[0177] A thickness of the optical adjustment layer **31** varies depending on the sub-pixels **101R**, **101G**, and **101B**. As the optical adjustment layer **31** has a thickness that varies depending on the sub-pixels **101R**, **101G**, and **101B**, it is possible to set optical distances for causing resonance suitable for the sub-pixels **101R**, **101G**, and **101B**.

[0178] In the example of FIG. **18A**, the positions of the first surfaces of the reflectors **30** provided in the sub-pixels **101R**, **101G**, and **101B** are arranged so as to be aligned in the up-down direction. In the sub-pixels **101R**, **101G**, and **101B**, the positions of the first surfaces of the second electrodes

15 vary with the differences in the thickness among the optical adjustment layers **31**.

[0179] The reflectors **30** can be formed with a metal such as aluminum (Al), silver (Ag), or copper (Cu), or an alloy containing these metals as principal components, for example.

[0180] The optical adjustment layers **31** can be formed with an inorganic insulating material such as silicon nitride (SiNx), silicon oxide (SiOx), or silicon oxynitride (SiOxNy), or an organic resin material such as an acrylic resin or a polyimide resin. Each of the optical adjustment layers **31** may be a single layer, or may be a multilayer film formed with a plurality of materials.

[0181] Each of the second electrodes **15** is preferably a layer that functions as a semi-transmissive reflective film. The second electrodes **15** can be formed with magnesium (Mg), silver (Ag), a magnesium-silver alloy (MgAg) containing these materials as the principal components, an alloy containing an alkali metal or an alkaline earth metal, or the like. The configurations of the first electrodes **13** and the organic layers **14** are similar to those described above, and therefore, explanation of them is not made herein.

(Resonator Structure: Second Example)

[0182] FIG. **18B** is a schematic cross-sectional view for explaining a second example in a case where the display device **10** has a resonator structure. The second example has a layer structure similar to that of the first example, except that the positions of the second electrodes **15** and the reflectors **30** are different from those in the first example.

[0183] In the sub-pixels **101R**, **101G**, and **101B** (light emitting elements **104R**, **104G**, and **104B**), the upper surfaces of the second electrodes **15** are arranged so that their positions in the up-down direction are aligned. The reflectors **30** provided in the sub-pixels **101R**, **101G**, and **101B** are at different positions in the up-down direction, depending on differences in thickness among the optical adjustment layers **31**.

(Resonator Structure: Third Example)

[0184] FIG. **19A** is a schematic cross-sectional view for explaining a third example in a case where the display device **10** has a resonator structure. The third example has a layer structure similar to that of the first example, except that the thicknesses of the reflectors **30** vary among the sub-pixels **101R**, **101G**, and **101B** (light emitting elements **104R**, **104G**, and **104B**).

[0185] In the sub-pixels **101R**, **101G**, and **101B**, the upper surfaces of the second electrodes **15** are arranged so that their positions in the up-down direction are aligned. The positions of the first surfaces of the reflectors **30** provided in the sub-pixels **101R**, **101G**, and **101B** vary in the up-down direction, depending on the differences in thickness among the optical adjustment layers **31**.

However, the positions of the second surfaces of the reflectors **30** are aligned among the sub-pixels **101R**, **101G**, and **101B**.

(Resonator Structure: Fourth Example)

[0186] FIG. **19B** is a schematic cross-sectional view for explaining a fourth example in a case where the display device **10** has a resonator structure. The fourth example is similar to the first example, except that the optical adjustment layers **31** are not included, and the thicknesses of the first electrodes **13** vary among the sub-pixels **101R**, **101G**, and **101B** (light emitting elements **104R**, **104G**, and **104B**).

[0187] Regarding the thicknesses of the first electrodes **13**, the respective thicknesses of the first electrodes **13** are designed so as to set optical distances for causing optical resonance suitable for the sub-pixels **101R**, **101G**, and **101B**.

(Resonator Structure: Fifth Example)

[0188] FIG. **20A** is a schematic cross-sectional view for explaining a fifth example in a case where the display device **10** has a resonator structure. The fifth example is similar to the first example, except that the optical adjustment layers **31** are not included, and oxide films **32** are formed on the first surface side (the side of the surfaces facing the first electrodes **13**) of the reflectors **30**.

[0189] Thicknesses of the oxide films **32** vary among the sub-pixels **101R**, **101G**, and **101B** (light emitting elements **104R**, **104G**, and **104B**).

[0190] Regarding the thicknesses of the oxide films **32**, the respective thicknesses of the oxide films **32** are designed so as to set optical distances for causing optical resonance suitable for the sub-pixels **101R**, **101G**, and **101B**.

[0191] The oxide films **32** are films obtained by oxidizing the surfaces of the reflectors **30**, and are formed with aluminum oxide, tantalum oxide, titanium oxide, magnesium oxide, zirconium oxide, or the like, for example. The oxide films **32** function as insulating films for adjusting the optical path lengths (optical distances) between the reflectors **30** and the second electrodes **15**.

[0192] The oxide films **32** having thicknesses suitable for the sub-pixels **101R**, **101G**, and **101B** can be formed in the following manner, for example.

[0193] First, a substrate on which the reflectors **30** are formed is immersed in a container filled with an electrolytic solution, and electrodes are disposed so as to face the reflectors **30**.

[0194] Then, with the electrodes being used as references, positive voltages are applied to the reflectors **30**, to anodize the reflectors **30**. Voltages corresponding to the thicknesses of the oxide films **32** to be obtained are applied to the reflectors **30** of the sub-pixels **101R**, **101G**, and **101B**. As a result, the oxide films **32** having different thicknesses (the oxide films **32** having thicknesses suitable for the sub-pixels **101R**, **101G**, and **101B**) can be collectively formed on the reflectors **30** of the sub-pixels **101R**, **101G**, and **101B**.

(Resonator Structure: Sixth Example)

[0195] FIG. **20B** is a schematic cross-sectional view for explaining a sixth example in a case where the display device **10** has a resonator structure.

[0196] In the sixth example, each resonator structure of the display device **10** is formed with a structure in which the first electrode **13**, the organic layer **14**, and the second electrode **15** are layered. In the sixth example, each first electrode **13** is a first electrode (also serving as a reflector) **33** that is designed to function as both an electrode and a reflector. The first electrodes (also serving as reflectors) **33** are formed with a material having an optical constant selected in accordance with the types of the light emitting elements **104R**, **104G**, and **104B**. Since the phase shift by the first electrodes (also serving as reflectors) **33** vary, it is possible to set an optical distance for generating optimum resonance for the wavelength of light corresponding to the color to be displayed.

[0197] The first electrodes (also serving as reflectors) **33** can be formed with a single-component metal such as aluminum (Al), silver (Ag), gold (Au), or copper (Cu), or an alloy containing these metals as the principal components. For example, a first electrode (also serving as a reflector) **33R** of the sub-pixel **101R** may be formed with copper (Cu), and a first electrode (also serving as a reflector) **33G** of the sub-pixel **101G** and a first electrode (also serving as a reflector) **33B** of the sub-pixel **101B** may be formed with aluminum.

[0198] The second electrodes **15** and the organic layers **14** are similar to those of the first example, and therefore, explanation of them is not made herein.

(Resonator Structure: Seventh Example)

[0199] FIG. **21** is a schematic cross-sectional view for explaining a seventh example in a case where the display device **10** has a resonator structure.

[0200] In the seventh example, the resonator structures illustrated in the sixth example are provided for the sub-pixels **101R** and **101G** (light emitting elements **104R** and **104G**), and the resonator structure illustrated in the first example is provided for the sub-pixel **101B** (light emitting element **104B**).

1-5 Fifth Embodiment

1-5-1 Configuration of Display Device

(Layout of Sub-Pixels)

[0201] A plurality of sub-pixels **101** in a display device **10** according to the fifth embodiment are two-dimensionally arranged in a stripe type layout as illustrated in FIG. **33**. Each of the sub-pixels **101** has, for example, a rectangular shape in plan view. A long side of the rectangular shape may be parallel to the Y axis.

[0202] The plurality of sub-pixels **101R** constitute a pixel column LR extending in the Y direction (longitudinal direction). The plurality of sub-pixels **101G** constitute a pixel column LG extending in the Y direction. The plurality of sub-pixels **101B** constitute a pixel column LB extending in the Y direction. The two pixel columns LG are arranged adjacent to each other in the X direction (lateral direction). The two pixel columns LR are arranged adjacent to each other in the X direction. One pixel column LB is arranged between two pixel columns LG and two pixel columns LR. Two pixel columns LR, one pixel column LB, two pixel columns LG, and one pixel column LB are repeatedly arranged in this order in the X direction. An arrangement pitch of the sub-pixels **101R** in the Y direction, an arrangement pitch of the sub-pixels **101G** in the Y direction, and an arrangement pitch of the sub-pixels **101B** in the Y direction may be the same or different. Hereinafter, an example in which these sub-pixels **101R**, **101G**, and **101B** are the same will be described.

[0203] A first block BK1 in which the pixel column LR, the pixel column LB, and the pixel column LG are arranged in this order in the X direction, and a second block BK2 in which the pixel column LG, the pixel column LB, and the pixel column LB are arranged in this order in the X direction are configured. The first block BK1 and the second block BK2 are alternately arranged in the X direction. The first block BK1 and the second block BK2 are symmetric with respect to an axis Ax. Here, the axis Ax is an axis passing between the first block BK1 and the second block BK2 and extending in the Y direction.

[0204] The sub-pixel **101G** included in one pixel column LG and the sub-pixel **101G** included in the other pixel column LG of the two adjacent pixel columns LG are arranged side by side in the X direction. The sub-pixel **101R** included in one pixel column LR and the sub-pixel **101R** included in the other pixel column LR of the two adjacent pixel columns LR are arranged in the X direction. The sub-pixel **101G** included in the pixel column LG and the sub-pixel **101R** included in the pixel column LR are arranged side by side in the X direction. The sub-pixel **101G** included in the pixel column LG and the sub-pixel **101B** included in the pixel column LB are arranged to be shifted in the Y direction. The shift amount is, for example, about $\frac{1}{2}$ of an arrangement pitch of the sub-pixels **101G** in the Y direction. The sub-pixel **101R** included in the pixel column LR and the sub-pixel **101B** included in the pixel column LB are arranged to be shifted in the Y direction. The shift amount is, for example, about $\frac{1}{2}$ of an arrangement pitch of the sub-pixels **101G** in the Y direction.

[0205] The sub-pixels **101G** and **101G** adjacent in the X direction are connected by a connection portion **23G1**. The sub-pixels **101G** and **101G** adjacent in the Y direction are connected by a connection portion **23G2**. The sub-pixels **101R** and **101R** adjacent in the X direction are connected by a connection portion **23R1**. The sub-pixels **101R** and **101R** adjacent in the Y direction are connected by a connection portion **23R2**. The sub-pixels **10G** and **10R** adjacent in the X direction with the sub-pixel **10B** interposed therebetween are connected by a connection portion **23RG**. The connection portion **23RG** passes between the sub-pixels **10B** and **10B** arranged in the Y direction. In the present specification, in a case where the connection portions **23G1**, **23G2**, **23R1**, **23R2**, and **23RG** are not particularly distinguished, the connection portions **23G1**, **23G2**, **23R1**, **23R2**, and **23RG** are collectively referred to as a connection portion **23**.

(Layer Configuration of Display Device)

[0206] As illustrated in FIGS. **34** to **37**, the display device **10** includes a drive substrate **11**, a plurality of first electrodes **13**, an organic layer **14G**, an organic layer **14R**, an organic layer **14B**, a second electrode **15**, a protective layer **61**, a sidewall **62**, an auxiliary electrode **63**, and a protective layer **64**.

(Organic Layer)

[0207] As illustrated in FIG. **38**, the organic layer **14G** includes a plurality of main bodies **14G0**, a plurality of coupling portions **14G1**, a plurality of coupling portions **14G2**, and a plurality of extension portions **14G3**. Each of the main bodies **14G0** is a portion constituting the sub-pixel **101G** (that is, a light emitting element **104G**) in the organic layer **14G**. Each of the coupling portions **14G1**, each of the coupling portions **14G2**, and each of the extension portions **14G3** are

arranged in an inter-sub-pixel region M. The coupling portion **14G1** extends in the lateral direction (+X direction and -X direction) from the main body **14G0**, and connects two main bodies **14G0** adjacent in the lateral direction. The coupling portion **14G2** extends in the longitudinal direction (+Y direction and -Y direction) from the main body **14G0**, and connects two main bodies **14G0** adjacent in the longitudinal direction. The extension portion **14G3** extends in the lateral direction (+X direction and -X direction) from the main body **14G0**, and a tip of the extension portion **14G3** is located between two main bodies **14G0** and **14R0** adjacent in the lateral direction with the organic layer **14B** interposed therebetween.

[0208] As illustrated in FIG. 38, the organic layer **14R** includes a plurality of main bodies **14R0**, a plurality of coupling portions **14R1**, a plurality of coupling portions **14R2**, and a plurality of extension portions **14R3**. Each of the main bodies **14R0** is a portion constituting the sub-pixel **101R** (that is, a light emitting element **104R**) in the organic layer **14R**. Each of the coupling portions **14R1**, each of the coupling portions **14R2**, and each of the extension portions **14R3** are arranged in the inter-sub-pixel region M. The coupling portion **14R1** extends in the lateral direction (+X direction and -X direction) from the main body **14R0** and connects two main bodies **14R0** adjacent in the lateral direction. The coupling portion **14R2** extends in the longitudinal direction (+Y direction and -Y direction) from the main body **14G0**, and connects two main bodies **14R0** adjacent in the longitudinal direction. The extension portion **14R3** extends in the lateral direction (+X direction and -X direction) from the main body **14R0**, and a tip of the extension portion **14R3** is located between two main bodies **14R0** and **14G0** adjacent in the lateral direction with the organic layer **14B** interposed therebetween.

[0209] As illustrated in FIG. 38, the organic layer **14B** includes a plurality of main bodies **140B**. In the fifth embodiment, an example in which the organic layer **14B** does not include an extension portion extending in a predetermined direction from the main body **14R0** will be described. However, the present disclosure is not limited to this example, and the organic layer **14B** may include an extension portion extending in a predetermined direction from the main body **14R0**.
(Second Electrode)

[0210] As illustrated in FIG. 39, the second electrode **15** includes a plurality of main bodies **15M0**, a plurality of coupling portions **15M1**, and a plurality of coupling portions **15M2**. Each of the main bodies **15M0** is a portion constituting the sub-pixel **101R** (light emitting element **104R**) or the sub-pixel **101G** (light emitting element **104G**) in the second electrode **15**. Each of the coupling portions **15M1** and each of the coupling portions **15M2** are arranged in the inter-sub-pixel region M. The coupling portion **15M1** extends in the lateral direction (+X direction and -X direction) from the main body **15M0**, and connects two main bodies **15M0** adjacent in the lateral direction. The coupling portion **15M2** extends in the longitudinal direction (+Y direction and -Y direction) from the main body **15M0**, and connects two main bodies **15M0** adjacent in the longitudinal direction. Each sub-pixel **101R** and each sub-pixel **101G** are electrically connected to each other by the coupling portion **15M1** and the coupling portion **15M2**, whereas each sub-pixel **101B** is isolated without being electrically connected to each other.

(Light Emitting Elements)

[0211] The sub-pixel **101R** includes the light emitting element **104R**. The light emitting element **104R** includes the first electrode **13**, the main body **14R0** of the organic layer **14R**, and the main body **15M0** of the second electrode **15**. The first electrode **13**, the main body **14R0** of the organic layer **14R**, and the main body **15M0** of the second electrode **15** are layered in this order on the first surface of the drive substrate **11**.

[0212] The sub-pixel **101G** includes the light emitting element **104G**. The light emitting element **104G** includes the first electrode **13**, the main body **14G0** of the organic layer **14G**, and the main body **15M0** of the second electrode **15**. The first electrode **13**, the main body **141G** of the organic layer **14G**, and the main body **15M0** of the second electrode **15** are layered in this order on the first surface of the drive substrate **11**.

[0213] The sub-pixel **101B** includes a light emitting element **104B**. The light emitting element **104B** includes the first electrode **13**, the main body **14B0** of the organic layer **14B**, and the main body **15M0** of the second electrode **15**. The first electrode **13**, the main body **141B** of the organic layer **14B**, and the main body **15M0** of the second electrode **15** are layered in this order on the first surface of the drive substrate **11**.

[0214] The light emitting elements **104R**, **104G**, and **104B** are two-dimensionally arranged on the first surface of the drive substrate **11** in an arrangement similar to an arrangement of the sub-pixels **101R**, **101G**, and **101B** described above.

(Connection Portion)

[0215] The connection portion **23G1** includes the coupling portion **14G1** of the organic layer **14G** and the coupling portion **15M1** of the second electrode **15**. The coupling portion **14G1** of the organic layer **14G** and the coupling portion **15M1** of the second electrode **15** are layered in this order on the first surface of the drive substrate **11**. The connection portion **23G2** includes the coupling portion **14G2** of the organic layer **14G** and the coupling portion **15M2** of the second electrode **15**. The coupling portion **14G2** of the organic layer **14G** and the coupling portion **15M2** of the second electrode **15** are layered in this order on the first surface of the drive substrate **11**.

[0216] The connection portion **23R1** includes the coupling portion **14R1** of the organic layer **14R** and the coupling portion **15M1** of the second electrode **15**. The coupling portion **14R1** of the organic layer **14R** and the coupling portion **15M1** of the second electrode **15** are layered in this order on the first surface of the drive substrate **11**. The connection portion **23R2** includes the coupling portion **14R2** of the organic layer **14R** and the coupling portion **15M2** of the second electrode **15**. The coupling portion **14R2** of the organic layer **14R** and the coupling portion **15M2** of the second electrode **15** are layered in this order on the first surface of the drive substrate **11**.

[0217] The connection portion **23RG** includes the extension portion **14G3** of the organic layer **14G**, the extension portion **14R3** of the organic layer **14R**, and the coupling portion **15M1** of the second electrode **15**. The extension portion **14G3** and the extension portion **14R3** are provided on the first surface of the drive substrate **11**. The extension portion **14G3** and the extension portion **14R3** may partially overlap each other, or the extension portion **14G3** and the extension portion **14R3** may be separated from each other. In a case where the extension portion **14G3** and the extension portion **14R3** partially overlap each other, the extension portion **14G3** may be located on an upper side of the extension portion **14R3**, or the extension portion **14R3** may be located on an upper side of the extension portion **14G3**. The coupling portion **15M1** is provided on the first surfaces of the extension portion **14G3** and the extension portion **14R3**.

(Protective Layer)

[0218] The protective layer **61** is provided on the first surface of the second electrode **15**. The protective layer **61** has translucency with respect to light emitted from the light emitting element **104**. The protective layer **61** can protect a plurality of the light emitting elements **104**, a plurality of the connection portions **23**, and the like. For example, the protective layer **61** can suppress moisture ingress into the plurality of light emitting elements **104** and the connection portions **23** from an external environment.

(Sidewall)

[0219] The sidewall **62** covers a side surface of each of the light emitting elements **104**. The sidewall **62** may further cover a side surface of the connection portion **23**. The sidewall **62** may have translucency with respect to light emitted from the light emitting element **104**. The sidewall **62** can protect the light emitting element **104**. For example, the sidewall **62** can suppress moisture ingress into the plurality of light emitting elements **104** from an external environment. As a material of the sidewall **62**, a material similar to that of the first protective layer **16** in the first embodiment can be exemplified.

(Auxiliary Electrode)

[0220] The auxiliary electrode **63** is provided on the first surface of the protective layer **61**. The

auxiliary electrode **63** includes a plurality of connection portions **631**. Each of the connection portions **631** is provided in a hole **611**, and connects the auxiliary electrode **63** to the first surface of the light emitting element **104B**, specifically, the first surface of the main body **15M0** of the second electrode **15**. The auxiliary electrode **63** is preferably a transparent electrode having transparency to visible light. Here, the transparent electrode includes, for example, a single-layer film of a metal layer, a single-layer film of a transparent conductive oxide layer, or a multilayer film of a metal layer and a transparent conductive oxide layer.

[0221] The metal layer contains, for example, at least one metal element selected from a group including magnesium (Mg), aluminum (Al), silver (Ag), calcium (Ca), and sodium (Na). The metal layer may contain the at least one metal element described above as a constituent element of an alloy. A specific example of the alloy includes an MgAg alloy, an MgAl alloy, an AlLi alloy, or the like.

[0222] The transparent conductive oxide layer contains a transparent conductive oxide. The transparent conductive oxide contains, for example, at least one selected from a group including an indium-containing transparent conductive oxide (hereinafter, referred to as an “indium-based transparent conductive oxide”), a tin-containing transparent conductive oxide (hereinafter, referred to as a “tin-based transparent conductive oxide”), and a zinc-containing transparent conductive oxide (hereinafter, referred to as a “zinc-based transparent conductive oxide”).

[0223] The indium-based transparent conductive oxide includes, for example, indium tin oxide (ITO), indium zinc oxide (IZO), indium gallium oxide (IGO), indium gallium zinc oxide (IGZO) or fluorine-doped indium oxide (IFO). The tin-based transparent conductive oxide contains, for example, tin oxide, antimony-doped tin oxide (ATO), or fluorine-doped tin oxide (FTO). The zinc-based transparent conductive oxide contains, for example, zinc oxide, aluminum-doped zinc oxide (AZO), boron-doped zinc oxide, or gallium-doped zinc oxide (GZO).

(Protective Layer)

[0224] The protective layer **64** is provided on the first surface of the auxiliary electrode **63**. The protective layer **64** has translucency with respect to light emitted from the light emitting element **104**. The protective layer **64** can protect a plurality of the light emitting elements **104**, the plurality of connection portions **23**, and the like. For example, the protective layer **61** can suppress moisture ingress into the plurality of light emitting elements **104** and the plurality of connection portions **23** from an external environment. As a material of the protective layer **61**, a material similar to that of the first protective layer **16** in the first embodiment can be exemplified.

(Contact Electrode)

[0225] The display device **10** may further include a contact electrode (not illustrated). The contact electrode is provided on the first surface of the drive substrate **11** around the display region **10A**. A peripheral edge portion of the second electrode **15** is connected to the contact electrode. The contact electrode is an auxiliary electrode that connects the second electrode **15** and a wiring (not illustrated) in the drive substrate **11**.

[0226] The contact electrode may have a closed loop shape surrounding the entire outer periphery of the display region **10A** in plan view, or may have a loop shape surrounding the outer periphery of the display region **10A** and divided at one location or a plurality of locations.

[0227] The contact electrode includes, for example, at least one of a metal layer and a metal oxide layer. More specifically, for example, the contact electrode includes a single-layer film of a metal layer or a metal oxide layer, or a multilayer film of a metal layer and a metal oxide layer. The contact electrode preferably has a configuration similar to that of the first electrode **13** described above. In this case, since the first electrode **13** and the contact electrode can be formed in the same step, the manufacturing process of the display device **10** can be simplified.

1-5-2 Operation and Effect

[0228] In the display device **10** according to the fifth embodiment described above, the connection portion **23G1** connecting the sub-pixels **101G** and **101G** adjacent in the X direction includes the

coupling portion **14G1** of the organic layer **14G** and the coupling portion **15M1** of the second electrode **15**. The connection portion **23G2** connecting the sub-pixels **101G** and **101G** adjacent in the Y direction includes the coupling portion **14G2** of the organic layer **14G** and the coupling portion **15M2** of the second electrode **15**. Therefore, the sub-pixel **10G** and the connection portions **23G1** and **23G2** can be formed in the same step. Therefore, the connection portions **23G1** and **23G2** connecting the sub-pixels **101G** and **101G** of the same color can be formed by a simple process.

[0229] Furthermore, the connection portion **23R1** connecting the sub-pixels **101R** and **101R** adjacent in the X direction includes the coupling portion **14R1** of the organic layer **14R** and the coupling portion **15M1** of the second electrode **15**. The connection portion **23R2** connecting the sub-pixels **101R** and **101R** adjacent in the Y direction includes the coupling portion **14R2** of the organic layer **14R** and the coupling portion **15M2** of the second electrode **15**. Therefore, the sub-pixel **10R** and the connection portions **23R1** and **23R2** can be formed in the same step. Therefore, the connection portions **23R1** and **23R2** connecting the sub-pixels **101R** and **101R** of the same color can be formed by a simple process.

[0230] In the display device **10** according to the fifth embodiment described above, since the two sub-pixels **101G** and **101G** having the same emission color are adjacent in the X direction, after one organic layer **14G** is formed so as to cover the two adjacent first electrodes **13**, the organic layer **14G** can be separated for each of the sub-pixels **101G** by a photolithography method. Similarly, since the two sub-pixels **101R** and **101R** having the same emission color are adjacent in the X direction, after one organic layer **14R** is formed so as to cover the two adjacent first electrodes **13**, the organic layer **14R** can be separated for each of the sub-pixels **101R** by the photolithography method. Therefore, the accuracy of a deposition mask for forming the organic layer **14R** and a deposition mask for forming the organic layer **14G** can be relaxed, and high definition of the display device **10** can be realized.

1-5-3 Modifications

(Modification 1)

[0231] In the fifth embodiment, an example in which the sub-pixel **101G** included in one pixel column LG and the sub-pixel **101G** included in the other pixel column LG of the two adjacent pixel columns LG are arranged so as to be aligned in the X direction has been described (see FIG. 33). However, the arrangement of the sub-pixels **101G** is not limited to this example. For example, as illustrated in FIG. 40, the sub-pixel **101G** included in one pixel column LG and the sub-pixel **101G** included in the other pixel column LG of the two adjacent pixel columns LG may be arranged to be shifted in the Y direction. The shift amount is, for example, about $\frac{1}{2}$ of an arrangement pitch of the sub-pixels **101G** in the Y direction.

(Modification 2)

[0232] In the fifth embodiment, an example in which the plurality of sub-pixels **101** is two-dimensionally arranged in a stripe type layout has been described (see FIG. 33). However, the layout of the plurality of sub-pixels **101** is not limited to this example. For example, the plurality of sub-pixels **101** may be two-dimensionally arranged in a delta type or square type layout. These layouts will be described below.

(Delta Type First Layout)

[0233] FIG. 41A is a plan view of a delta type first layout. A pixel column LR, a pixel column LB, a pixel column LG, and a pixel column LB are repeatedly arranged in this order in the X direction. The pixel column LR is configured by arranging a plurality of sub-pixels **101R** in a staggered manner. The pixel column LG is configured by arranging a plurality of sub-pixels **101G** in a staggered manner. The pixel column LB is configured by arranging a plurality of sub-pixels **101B** in a staggered manner.

[0234] Each of the sub-pixels **101G** includes two sub-pixel elements **101G1** and **101G2** adjacent in the X direction. Each of the sub-pixels **101R** includes two sub-pixel elements **101R1** and **101R2**

adjacent in the X direction.

[0235] The sub-pixel elements **101R1** and **101R2** adjacent in the X direction are connected by a connection portion **23R1**. The sub-pixel elements **101G1** and **101G2** adjacent in the X direction are connected by a connection portion **23G1**. The sub-pixel elements **101R1** and **101R2** adjacent in the Y direction are connected by a connection portion **23R2**. The sub-pixel elements **101G1** and **101G2** adjacent in the Y direction are connected by a connection portion **23G2**.

[0236] The sub-pixel elements **101R2** and **101G1** adjacent in an oblique direction with the sub-pixel **101B** interposed therebetween are connected by a connection portion **23RG**. The sub-pixel elements **101G2** and **101R1** adjacent in the oblique direction with the sub-pixel **101B** interposed therebetween are connected by a connection portion **23RG**. Here, the oblique direction represents a direction between the X direction and the Y direction and a direction between the X direction and the -Y direction.

[0237] The sub-pixels **101R**, **101G**, and **101B** have, for example, a substantially hexagonal shape in plan view. The sub-pixel elements **101R1** and **101R2** and the sub-pixel elements **101G1** and **101G2** have, for example, a substantially trapezoidal shape in plan view. The sub-pixel elements **101R1** and **101R2** are arranged such that lower bases of the substantially trapezoidal shapes face each other in plan view. The sub-pixel elements **101R1** and **101R2** are arranged such that lower bases of the substantially trapezoidal shapes face each other in plan view.

(Delta Type Second Layout)

[0238] FIG. **41B** is a plan view of a delta type second layout. The delta type second layout is different from the delta type first layout in that sub-pixels **101G** and **101R** have a substantially circular shape in plan view. Sub-pixel elements **101R1** and **101R2** and sub-pixel elements **101G1** and **101G2** have, for example, a substantially semicircular shape in plan view. The sub-pixel elements **101R1** and **101R2** are arranged such that their substantially semicircular strings face each other. The sub-pixel elements **101G1** and **101G2** are arranged such that their substantially semicircular strings face each other.

(Delta Type Third Layout)

[0239] FIG. **42A** is a plan view of a delta type third layout. A pixel column LR, a pixel column LG, and a pixel column LB are repeatedly arranged in this order in the X direction. In this case, the pixel column LR is configured by linearly arranging a plurality of sub-pixels **101R**. The pixel column LG is configured by linearly arranging a plurality of sub-pixels **101G**. The pixel column LB is configured by linearly arranging a plurality of sub-pixels **101B**.

[0240] The sub-pixel elements **101R1** and **101R2** adjacent in the X direction are connected by a connection portion **23R1**. The sub-pixel elements **101G1** and **101G2** adjacent in the X direction are connected by a connection portion **23G1**.

[0241] The connection portions **23R1** of the sub-pixels **101R** and **101R** adjacent in the Y direction are connected by a connection portion **23R2**. The connection portions **23G1** of the sub-pixels **101G** and **101G** adjacent in the Y direction are connected by a connection portion **23G2**.

[0242] The sub-pixel elements **101R2** and **101G1** adjacent in the X direction with the sub-pixel **101B** interposed therebetween are connected by a connection portion **23RG**. The sub-pixel elements **101G2** and **101R1** adjacent in the X direction with the sub-pixel **101B** interposed therebetween are connected by a connection portion **23RG**.

[0243] The sub-pixels **101R**, **101G**, and **101B** have, for example, a substantially hexagonal shape in plan view. The configurations of the sub-pixels **101R**, **101G**, and **101B** are similar to those of the delta type first layout.

(Delta Type Fourth Layout)

[0244] FIG. **42B** is a plan view of a delta type fourth layout. The delta type fourth layout is different from the delta type third layout in that sub-pixels **101G** and **101R** have a substantially rhombic shape in plan view. Sub-pixel elements **101R1** and **101R2** and sub-pixel elements **101G1** and **101G2** have, for example, a substantially triangular shape in plan view. The sub-pixel elements

101R1 and **101R2** are arranged such that sides of the substantially triangular shapes face each other. The sub-pixel elements **101G1** and **101G2** are arranged such that their substantially triangular sides face each other.

[0245] Note that the shapes of the sub-pixels **101R**, **101G**, and **101B** in plan view are not limited to the above-described substantially hexagonal shape, substantially circular shape, and substantially rhombic shape, and may be shapes other than these shapes. For example, the shapes of the sub-pixels **101G** and **101R** may have a substantially elliptical shape or a substantially polygonal shape other than a substantially rhombic shape in plan view. The shapes of the sub-pixel elements **101R1**, **101R2**, **101G1**, and **101G2** in a plan view are not limited to the above-described substantially trapezoidal shape, substantially circular shape, and substantially triangular shape, and may be other shapes. For example, the shape of each of the sub-pixel elements **101R1**, **101R2**, **101G1**, and **101G2** may be a substantially polygonal shape such as a substantially semi-elliptical shape or a substantially quadrangular shape.

(Square Type Layout)

[0246] FIG. 43 is a plan view of a square type layout. One pixel includes a sub-pixel **101R**, a sub-pixel **101G**, a sub-pixel **10B**, and a sub-pixel **10B** two-dimensionally arranged in a square layout. A pixel column LGB, a pixel column LGB, a pixel column LRB, and a pixel column LRB are repeatedly arranged in this order in the X direction. The pixel column LGB is configured by repeatedly arranging the sub-pixel **10G**, the sub-pixel **10G**, the sub-pixel **10B**, and the sub-pixel **10B** in this order in the Y direction. The pixel column LRB is configured by repeatedly arranging the sub-pixel **10R**, the sub-pixel **10R**, the sub-pixel **10B**, and the sub-pixel **10B** in this order in the Y direction.

[0247] The sub-pixel **101G** included in one pixel column LGB and the sub-pixel **101G** included in the other pixel column LGB of the two adjacent pixel columns LGB and LGB are arranged side by side in the X direction. The sub-pixel **101B** included in one pixel column LGB and the sub-pixel **101B** included in the other pixel column LGB of the two adjacent pixel columns LGB and LGB are arranged side by side in the X direction.

[0248] The sub-pixel **101R** included in one pixel column LRB and the sub-pixel **101R** included in the other pixel column LRB of the two adjacent pixel columns LRB and LRB are arranged in the X direction. The sub-pixel **101B** included in one pixel column LRB and the sub-pixel **101B** included in the other pixel column LRB of the two adjacent pixel columns LRB and LRB are arranged in the X direction.

[0249] Of the two adjacent pixel columns LGB and LRB, the sub-pixel **101G** included in the pixel column LGB and the sub-pixel **101B** included in the pixel column LRB are arranged side by side in the X direction. Of the two adjacent pixel columns LGB and LRB, the sub-pixel **101B** included in the pixel column LGB and the sub-pixel **101R** included in the pixel column LRB are arranged side by side in the X direction.

[0250] The two sub-pixels **101G** arranged in the X direction are connected by a connection portion **23G1**. The two sub-pixels **101R** arranged in the X direction are connected by a connection portion **23R1**. The sub-pixel **101R** and the sub-pixel **101G** adjacent in the oblique direction are connected by a connection portion **23RG**.

(Modification 3)

[0251] In the fifth embodiment, an example in which the sub-pixels **101G** and **101G** adjacent in the X direction and the sub-pixels **101G** and **101G** adjacent in the Y direction are connected by the different connection portions **23G1** and **23G2** has been described (see FIG. 33). However, the connection form of the sub-pixels **101G** and **101G** adjacent in the X direction and the sub-pixels **101G** and **101G** adjacent in the Y direction is not limited to this example. For example, as illustrated in FIG. 52, sub-pixels **101G** and **101G** adjacent in the X direction and sub-pixels **101G** and **101G** adjacent in the Y direction may be connected by one connection portion **23G3**. A plurality of sub-pixels **101G** included in the two pixel columns LG may be connected by one

connection portion **23G3**. The connection portion **23G3** is provided in an inter-sub-pixel region M. [0252] Similarly to the sub-pixels **101G** and **101G** adjacent in the X direction and the sub-pixels **101G** and **101G** adjacent in the Y direction, as illustrated in FIG. 52, the sub-pixels **101R** and **101R** adjacent in the X direction and the sub-pixels **101R** and **101R** adjacent in the Y direction may also be connected by one connection portion **23R3**. A plurality of sub-pixels **101R** included in the two pixel columns LR may be connected by one connection portion **23R3**. The connection portion **23R3** is provided in the inter-sub-pixel region M.

[0253] However, while the sub-pixels **101G** and **101G** adjacent in the X direction and the sub-pixels **101G** and **101G** adjacent in the Y direction are connected by one connection portion **23G3**, the sub-pixels **101R** and **101R** adjacent in the X direction and the sub-pixels **101R** and **101R** adjacent in the Y direction may be connected by different connection portions **23R1** and **23R2**. Conversely, while the sub-pixels **101R** and **101R** adjacent in the X direction and the sub-pixels **101R** and **101R** adjacent in the Y direction are connected by one connection portion **23R3**, the sub-pixels **101G** and **101G** adjacent in the X direction and the sub-pixels **101G** and **101G** adjacent in the Y direction may be connected by different connection portions **23G1** and **23G2**.

[0254] FIG. 53 is a plan view of organic layers **14R**, **14G**, and **14B** in a case where the display device **10** includes connection portions **23G3** and **23R3**. The organic layer **14G** includes a plurality of main bodies **14GL** and a plurality of extension portions **14G3**. Each of the main bodies **14GL** is a portion constituting a plurality of sub-pixels **101G** included in the two pixel columns LG and a plurality of connection portions **23G3** connecting the sub-pixels **101G** to each other. The plurality of main bodies **14GL** has a stripe shape. Each of the extension portions **14G3** is a portion constituting a connection portion **23RG** that connects the sub-pixels **101G** and **101R** adjacent in the lateral direction with the organic layer **14B** interposed therebetween. The extension portion **14G3** extends in the lateral direction (+X direction and -X direction) from the main body **14GL**, and a tip of the extension portion **14G3** is located between the main bodies **14GL** and **14RL** adjacent in the lateral direction with the organic layer **14B** interposed therebetween.

[0255] The organic layer **14R** includes a plurality of main bodies **14RL** and a plurality of extension portions **14R3**. Each of the main bodies **14RL** is a portion constituting a plurality of sub-pixels **101R** included in the two pixel columns LR and a plurality of connection portions **23R3** connecting the sub-pixels **101R** to each other. The plurality of main bodies **14RL** has a stripe shape. Each of the extension portions **14R3** is a portion constituting a connection portion **23RG** that connects the sub-pixels **101R** and **101G** adjacent in the lateral direction with the organic layer **14B** interposed therebetween. The extension portion **14R3** extends in the lateral direction (+X direction and -X direction) from the main body **14RL**, and a tip of the extension portion **14R3** is located between the main bodies **14RL** and **14GL** adjacent in the lateral direction with the organic layer **14B** interposed therebetween.

[0256] FIG. 54 is a plan view of a second electrode **15** in a case where the display device **10** includes connection portions **23G3** and **23R3**. The second electrode **15** includes a plurality of main bodies **15ML**, a plurality of main bodies **15M0**, and a plurality of coupling portions **15M1**. Each of the main bodies **15ML** is a portion constituting a plurality of sub-pixels **101G** included in two pixel columns LG, a plurality of connection portions **23G3** connecting the plurality of sub-pixels **101G** to each other, a plurality of sub-pixels **101R** included in two pixel columns LR, and a plurality of connection portions **23R3** connecting the plurality of sub-pixels **101R** to each other. The plurality of main bodies **15ML** has a stripe shape. Each of the coupling portions **15M1** is a portion constituting a connection portion **23RG** that connects the sub-pixels **101R** and **101G** adjacent in the lateral direction with a sub-pixel **101B** interposed therebetween. The coupling portion **15M1** extends in the lateral direction (+X direction and -X direction) from the main body **15ML** and connects the main bodies **15ML** adjacent in the lateral direction.

(Modification 4)

[0257] In the fifth embodiment, an example in which the organic layer **14G** includes the coupling

portion **14G1**, the coupling portion **14G2**, and the extension portion **14G3** has been described. However, the configuration of the organic layer **14G** is not limited thereto. For example, the organic layer **14G** may not include at least one of the coupling portion **14G1**, the coupling portion **14G2**, and the extension portion **14G3**. The organic layer **14G** may include an extension portion extending in the lateral direction (+X direction and -X direction) from the main body **14G0** instead of the coupling portion **14G1**. A tip of the extension portion is located between two main bodies **14G0** adjacent in the lateral direction. The organic layer **14G** may include an extension portion extending in the longitudinal direction (+Y direction and -Y direction) from the main body **14G0** instead of the coupling portion **14G2**. A tip of the extension portion is located between two main bodies **14G0** adjacent in the longitudinal direction.

[0258] In the fifth embodiment, an example in which the organic layer **14R** includes the coupling portion **14R1**, the coupling portion **14R2**, and the extension portion **14R3** has been described. However, the configuration of the organic layer **14R** is not limited thereto. For example, the organic layer **14R** may not include at least one of the coupling portion **14R1**, the coupling portion **14R2**, and the extension portion **14R3**. The organic layer **14R** may include an extension portion extending in the lateral direction (+X direction and -X direction) from the main body **14R0** instead of the coupling portion **14R1**. A tip of the extension portion is located between two main bodies **14R0** adjacent in the lateral direction. The organic layer **14R** may include an extension portion extending in the longitudinal direction (+Y direction and -Y direction) from the main body **14R0** instead of the coupling portion **14R2**. A tip of the extension portion is located between two main bodies **14R0** adjacent in the longitudinal direction.

(Modification 5)

[0259] In the fifth embodiment, an example in which the two sub-pixels **101G** of the same emission color are adjacent in the X direction and the two sub-pixels **101R** of the same emission color are adjacent in the X direction has been described. However, the layout of the sub-pixel **101G** and the sub-pixel **101R** is not limited thereto. For example, only two sub-pixels of one of the sub-pixel **101G** and the sub-pixel **101G** may be arranged adjacent to each other.

(Modification 6)

[0260] The display device **10** according to the fifth embodiment may have a resonator structure in at least a part of the plurality of sub-pixels **101**. The resonator structure is as described in the fourth embodiment.

2 Method of Manufacturing Display Device

2-1-1 Contents of Manufacturing Method

[0261] A method of manufacturing a light emitting device will be described with reference to FIGS. **7** to **13**, taking as an example a case where the light emitting device is the display device **10** described in the above first embodiment. However, here, the description will be continued by taking, as an example, a case where the hole injection layer **140**, the electron transport layer **143**, and the electron injection layer **144** in the layer (functional layer **25**) excluding the light emitting layer **142** in the organic layer **14** are common to the plurality of sub-pixels **101**, and the hole transport layer **141** is different. Note that FIGS. **7** to **9** are cross-sectional views (process cross-sectional views) schematically illustrating a manufacturing process at a position corresponding to the cross section taken along line I-I in FIG. **2**. FIG. **10** is a cross-sectional view (process cross-sectional view) schematically illustrating a manufacturing process at a position corresponding to the cross section taken along line II-II in FIG. **2**. FIGS. **11** to **13** are cross-sectional views (process cross-sectional views) schematically illustrating a manufacturing process at a position corresponding to a cross section taken along line III-III in FIG. **2**. Note that the longitudinal cross section taken along line II-II in FIG. **2** is formed similarly to the longitudinal cross section taken along line I-I in FIG. **2** until immediately before the resist is provided, and thus illustration is omitted.

(Formation Step of First Electrode)

[0262] As illustrated in FIGS. 7A and 11A, the first electrode **13** is patterned on the drive substrate **11** provided with a circuit, a contact plug, and the like according to the layout of the sub-pixel **101**. The insulating layer **12** is patterned between the adjacent first electrodes **13**. The opening **12A** is formed in the insulating layer **12**, and the first electrode **13** is exposed from the opening **12A**.

(First Step)

[0263] The organic layer **14** including the light emitting layer is patterned on the first electrode **13** using a mask determined according to the layout of the plurality of sub-pixels **101**. This step is referred to as a first step.

[0264] In the first step, for example, as illustrated in FIGS. 7B and 11B, the first layer **125A** (the hole injection layer **140** and the hole transport layer **141**) corresponding to the organic layer **14B** of the sub-pixel **101B** is formed. Moreover, as illustrated in FIGS. 7C and 11C, a mask **150** is disposed above the first layer **125A** to form the light emitting layer **142** (light emitting layer **142B**). Here, the mask **150** is a mask **150B** corresponding to the sub-pixel **101B**. The mask **150** is determined for each color type of the plurality of sub-pixels **101** corresponding to each of the plurality of emission colors, and masks **150B**, **150G**, and **150R** corresponding to the respective sub-pixels **101B**, **101G**, and **101R** are prepared.

[0265] More specifically, for example, the mask **150B** includes a plurality of openings arranged in an arrangement pattern similar to that of the plurality of sub-pixels **101B**. When the organic layer **14B** is formed, the mask **150B** is disposed to face the first surface of the drive substrate **11** such that each opening is located above the first electrode **13** of the sub-pixel **10B**. The mask **150G** includes a plurality of openings arranged in an arrangement pattern similar to the plurality of sub-pixels **101G**. When the organic layer **14G** is formed, the mask **150G** is disposed to face the first surface of the drive substrate **11** such that each opening is located above the first electrode **13** of the sub-pixel **10G**. The mask **150R** includes a plurality of openings arranged in an arrangement pattern similar to the plurality of sub-pixels **101R**. When the organic layer **14R** is formed, the mask **150R** is disposed to face the first surface of the drive substrate **11** such that each opening is located above the first electrode **13** of the sub-pixel **10R**.

[0266] Next, as illustrated in FIGS. 8A and 12A, the mask **150B** is changed to the mask **150G** corresponding to the sub-pixel **101G**, and the light emitting layer **142G** corresponding to the sub-pixel **101G** is formed. Note that, in this case, the hole transport layer **141** is preferably further additionally formed so as to have a thickness suitable for the sub-pixel **101G**. In this case, after the mask **150G** is disposed, the hole transport layer **141** is additionally formed, and the light emitting layer **142G** corresponding to the sub-pixel **101G** is further formed.

[0267] Moreover, as illustrated in FIGS. 8B and 12B, the mask **150G** is changed to the mask **150R** corresponding to the sub-pixel **101R**, and the light emitting layer **142R** corresponding to the sub-pixel **101R** is formed. Note that, in this case, the hole transport layer **141** is preferably further additionally formed so as to have a thickness suitable for the sub-pixel **101R**. In this case, after the mask **150R** is disposed, the hole transport layer **141** is additionally formed, and the light emitting layer **142R** corresponding to the sub-pixel **101R** is further formed. As described above, in the case of forming the plurality of sub-pixels **101** corresponding to the plurality of emission colors, in the first step, it is preferable to change the mask **150** for each color type of the sub-pixel **101** to form the light emitting layer **142** corresponding to the plurality of sub-pixels **101**.

[0268] After the light emitting layer **142** corresponding to the sub-pixel **101** is formed, the second layer **125B** (the electron transport layer **143** and the electron injection layer **144**) is formed as illustrated in FIGS. 8C and 12C. In the examples of FIGS. 8C and 12C, a common layer is used for the second layer **125B** regardless of the sub-pixel **101**. However, this is an example, and the second layer **125B** may also include a layer having a different thickness or the like according to the color type of the sub-pixel **101**, similarly to the first layer **125A**.

(Second Step)

[0269] After the first step, a second step is performed. The second step is a step of layering the

second electrode **15** on the organic layer **14**. In the second step, as illustrated in FIGS. **8C** and **12C**, the second electrode **15** is formed on the entire surface on the first surface side. After the second electrode **15** is formed, a protective layer (first protective layer **16**) is formed.

(Third Step)

[0270] The third step is a step of removing, by etching, a portion of the organic layer **14** and the second electrode that is away from a portion combining the sub-pixel and the connection portion connecting the plurality of different sub-pixels.

[0271] In the third step, as illustrated in FIGS. **9A**, **10A**, and **13A**, a resist **151** is disposed on the first protective layer **16**. The resist **151** is formed in a pattern corresponding to a combined portion of the sub-pixel **101** and the connection portion **23**. Next, as illustrated in FIG. **9A**, portions of the first protective layer **16**, the second electrode **15**, and the organic layer **14** exposed without being covered with the resist **151** are removed by etching. As illustrated in FIGS. **10B** and **13B**, the first protective layer **16**, the second electrode **15**, and the organic layer **14** are left in the portion where the connection portion **23** is formed in addition to the portion corresponding to the sub-pixel **101**. At this time, as illustrated in FIGS. **10B** and **13B**, the side wall **24** of the multilayer structure **22** is preferably formed. Then, the resist **151** is removed, and the second protective layer **17** is further formed on one surface as illustrated in FIGS. **9C**, **10C**, and **13C**. Moreover, the low refractive index layer **18** and the like are formed as necessary. Thus, the display device **10** is obtained.

[0272] In a case where the mask **150** having a pattern corresponding to the layout of the sub-pixel **101** is disposed and the light emitting layer **142** is pattern-formed, it is preferable that the position of the mask **150** is accurately disposed at a position corresponding to the layout of the sub-pixel **101**, but the position of the mask **150** may be disposed at a position shifted within a predetermined range (within an allowable range) from the position corresponding to the layout of the sub-pixel **101**. Since the portion deviated from the combined portion of the sub-pixel **101** and the connection portion **23** is removed by etching, the portion deviated from the sub-pixel **101** and the connection portion **23** in the positional deviation portion of the light emitting layer **142** and the functional layer **25** caused by the positional deviation of the mask **150** is deleted. In the examples illustrated in FIGS. **7** to **13**, a resist covering the combined portion of the sub-pixel **101** and the connection portion **23** is formed using a photolithography method, and a portion deviated from the combined portion of the sub-pixel **101** and the connection portion **23** is removed using an etching method.

[0273] The mask used in the step of forming the organic layer **14** is not particularly limited as long as the layout of each layer such as the light emitting layer **142** constituting the organic layer **14** can be formed (pattern formation) in a desired size, and for example, fine metal mask (FMN), membrane mask, and the like can be exemplified.

[0274] The method of forming each layer of the light emitting layer **142** for forming the organic layer **14** and the other layers **126** excluding the light emitting layer is not particularly limited, and examples thereof include a vapor deposition method and a coating method.

[0275] Examples of the method of forming the second electrode **15** include a vapor deposition method and a sputtering method. Examples of the method of forming the first protective layer **16** and the second protective layer **17** include a vapor deposition method and a sputtering method, similarly to the method of forming the second electrode.

2-1-2 Operation and Effect

[0276] According to the above-described method of manufacturing a display device, even in a case where a plurality of sub-pixels having a plurality of emission colors and different emission colors is provided, it is not necessary to form a pattern of the second electrode for each type of sub-pixel, and thus, it is possible to realize facilitation of the manufacturing process. Furthermore, according to the method of manufacturing the display device described above, it is not necessary to be exposed to the atmosphere during the process of forming the organic layer **14** of each of the plurality of types of sub-pixels **101R**, **101G**, and **101B**, and it is possible to form the plurality of types of sub-pixels **101** in one vacuum state.

[0277] Furthermore, in the method of manufacturing the display device, since the light emitting layer **142** constituting the organic layer **14** is formed using the mask **150**, in a case where the positional deviation of the mask **150** occurs, a state in which the light emitting layers **142** of the adjacent sub-pixels **101** overlap in the inter-sub-pixel region M or a state in which the light emitting layer **142** extends to the outside of the sub-pixel **101** may be formed depending on the magnitude of the positional deviation. If such a state is formed in a wide range in the display region **10A**, the color gamut and resolution of the image displayed in the display region **10A** may be adversely affected. In this regard, in the above-described method of manufacturing a display device, a portion deviated from a portion corresponding to the sub-pixel **101** and the connection portion **23** is removed using a photolithography method and etching. As a result, the positions of the side end surfaces of the respective layers such as the light emitting layer **142** constituting the organic layer formed in each sub-pixel are aligned, and a state in which the light emitting layers **142** of the adjacent sub-pixels **101** overlap in the inter-sub-pixel region M in the portion excluding the connection portion **23** and a state in which the light emitting layer **142** extends to the outside of the sub-pixel **101** can be limited to a limited range. Therefore, it is possible to obtain a display device excellent in the color gamut and resolution of the image displayed in the display region **10A**.

2-1-3 Modification of Manufacturing Method

[0278] In the above description of the method of manufacturing the display device, the hole injection layer **140** is common to the plurality of types of sub-pixels **101R**, **101G**, and **101B**, but the manufacturing method is not limited thereto. The thickness and the like of the hole injection layer **140** may be different in the plurality of types of sub-pixels **101R**, **101G**, and **101B**. In that case, the mask **150** is disposed before forming the hole injection layer **140** of the organic layer **14** that forms the sub-pixel (for example, the sub-pixel **101B**) of the first color in the first step (modification). The case where the mask **150** is disposed before the hole injection layer **140** is formed in the first step in this manner is referred to as a modification of the manufacturing method. For example, in a modification of the manufacturing method, the first layer **125A** (hole injection layer **140** and hole transport layer **141**) and the light emitting layer **142** are formed in a state where the mask **150B** of the sub-pixel **101B** is disposed. Note that, in a case where the second layer **125B** (the electron transport layer **143** and the electron injection layer **144**) is also different in several types of sub-pixels **101R**, **101G**, and **101B**, the electron transport layer **143** and the electron injection layer **144** are formed in a state where the mask **150B** is continuously disposed.

[0279] Next, the mask **150B** is changed to a mask **150G**, and similarly to the sub-pixel **101B**, the first layer **125A** (hole injection layer **140** and hole transport layer **141**) and the light emitting layer **142** corresponding to the sub-pixel **101G** are formed. Moreover, in a case where the second layer **125B** (the electron transport layer **143** and the electron injection layer **144**) is different in several types of sub-pixels **101R**, **101G**, and **101B**, the electron transport layer **143** and the electron injection layer **144** corresponding to the sub-pixel **101G** are formed in a state where the mask **150G** is continuously disposed.

[0280] Note that, in a case where the second layer **125B** is common to the plurality of types of sub-pixels **101R**, **101G**, and **101B**, after the first layer **125A** and the light emitting layer **142** are formed for the sub-pixels **101R**, **101G**, and **101B**, the second layer **125B** may be formed in a state where the mask **150** is removed.

[0281] In the modification of the manufacturing method, the steps similar to those of the manufacturing method described above are performed for the other steps of the first step described above, and thus the description thereof will be omitted.

2-2-1 Contents of Manufacturing Method

[0282] Hereinafter, a method of manufacturing the display device **10** according to the fifth embodiment will be described with reference to FIGS. **44A** to **51D**. Reference signs R, G, and B described below the drive substrate **11** in FIGS. **44A** to **51D** represent formation positions of the

sub-pixels **10R**, **10G**, and **10B** in the X direction, respectively. FIGS. **44A** to **47D** are process diagrams corresponding to the cross section illustrated in FIG. **34** (cross section taken along line XXXIV-XXXIV in FIG. **33**). FIGS. **48A** to **51D** are process diagrams corresponding to the cross section illustrated in FIG. **35** (cross section taken along line XXXV-XXXV in FIG. **33**).

(Formation Step of First Electrode)

[0283] First, a metal layer and a metal oxide layer are sequentially formed on the first surface of the drive substrate **11** by, for example, a sputtering method, and then the metal layer and the metal oxide layer are patterned by, for example, a photolithography method. As a result, as illustrated in FIGS. **44A** and **48A**, the plurality of first electrodes **13** is formed on the first surface of the drive substrate **11**.

(Formation Step of Organic Layer Having Green Color as Emission Color)

[0284] Next, as illustrated in FIGS. **44B** and **48B**, the mask **71** is disposed to face above the first surface of the drive substrate **11** such that the opening **71A** of the mask **71** is located above the two columns of first electrodes **13** corresponding to the two adjacent pixel columns LG. At this time, the two columns of first electrodes **13** corresponding to the two adjacent pixel columns LG may be exposed from one opening **71A**, or the two adjacent sub-pixels **101G** may be exposed from one opening **71A**. Thereafter, the organic layer **14G** is formed on the first surface of the drive substrate **11** via the mask **71** by, for example, a vapor deposition method. As a result, the two columns of first electrodes **13** corresponding to the two adjacent pixel columns LG are covered with the organic layer **14G**.

[0285] When the organic layer **14G** is formed, the organic layer **14G** may be formed in a formation area of the organic layer **14B** (for example, on the first electrode **13** for forming the organic layer **14B**) due to formation variations, deposition blurring, and the like. Here, the formation variation represents variations due to the formation accuracy of the opening **71A** of the mask **71**, misalignment between the drive substrate **11** and the mask **71**, thermal expansion of the mask **71**, and the like. The deposition blur represents a phenomenon in which the boundary of the deposition pattern is blurred due to wraparound, vignetting, or the like of the deposition material.

[0286] The mask **71** has a plurality of openings **71A** two-dimensionally arranged in a stripe type layout. The openings **71A** may be arranged in the same pattern as the two adjacent pixel columns LG, or may be arranged in the same pattern as the two adjacent sub-pixels **101G**. A width of the opening **71A** in the X direction is, for example, about twice the arrangement pitch of the sub-pixels **101** in the X direction. An edge of the opening **71A** of the mask **71** is located, for example, between the first electrode **13** for forming the sub-pixel **101G** and the first electrode **13** for forming the sub-pixel **101B**. The mask **71** is, for example, a fine metal mask (FMM) or a membrane mask.

(Formation Step of Organic Layer Having Red Color as Emission Color)

[0287] Next, as illustrated in FIGS. **44C** and **48C**, the mask **72** is disposed to face above the first surface of the drive substrate **11** such that the opening **72A** of the mask **72** is located above the two columns of first electrodes **13** corresponding to the two adjacent pixel columns LR. At this time, two columns of first electrodes **13** corresponding to two adjacent pixel columns LR may be exposed from one opening **72A**, or two adjacent sub-pixels **101R** may be exposed from one opening **72A**. The mask **72** is disposed such that the first electrode **13** for forming one sub-pixel **101B** exists between the opening **71A** of the mask **71** and the opening **72B** of the mask **72**. Thereafter, the organic layer **14R** is formed on the first surface of the drive substrate **11** via the mask **72** by, for example, a vapor deposition method. As a result, the two columns of first electrodes **13** corresponding to the two adjacent pixel columns LR are covered with the organic layer **14R**.

[0288] When the organic layer **14R** is formed, there is a possibility that the organic layer **14R** is formed in a formation area of the organic layer **14B** (for example, on the first electrode **13** for forming the organic layer **14R**) due to formation variations, vapor deposition blurring, and the like similarly to the above organic layer **14G**.

[0289] The mask **72** has a plurality of openings **72A** two-dimensionally arranged in a stripe type layout. The openings **72A** may be arranged in the same pattern as two adjacent pixel columns **LR**, or may be arranged in the same pattern as two adjacent sub-pixels **101R**. A width of the opening **72A** in the X direction is, for example, about twice the arrangement pitch of the sub-pixels **101** in the X direction. An edge of the opening **72A** of the mask **72** is located, for example, between the first electrode **13** for forming the sub-pixel **101R** and the first electrode **13** for forming the sub-pixel **101B**. The mask **72** is, for example, a fine metal mask (FMM) or a membrane mask.

(Formation Step of Second Electrode)

[0290] Next, as illustrated in FIGS. **44D** and **48D**, the second electrode **15** is formed on the first surface of the organic layer **14R** and the first surface of the organic layer **14G** by, for example, a vapor deposition method or a sputtering method.

(Formation Step of Protective Layer)

[0291] Next, as illustrated in FIGS. **44D** and **48D**, a protective layer **61** is formed on the first surface of the second electrode **15** by, for example, a CVD method or a vapor deposition method.

(Patterning Step of Organic Layer, Second Electrode and Protective Layer)

[0292] Next, the protective layer **61**, the second electrode **15**, the organic layer **14G**, and the organic layer **14R** are patterned by, for example, a photolithography method. More specifically, as illustrated in FIGS. **44E** and **48E**, a photoresist layer **73** having a predetermined pattern is formed on the first surface of the protective layer **61**. Subsequently, as illustrated in FIGS. **45A** and **49A**, the protective layer **61**, the second electrode **15**, the organic layer **14G**, and the organic layer **14R** are processed by, for example, dry etching, and then the photoresist layer **73** is removed. As a result, the plurality of light emitting elements **104G**, the plurality of light emitting elements **104R**, the plurality of connection portions **23G1**, the plurality of connection portions **23G2**, the plurality of connection portions **23R1**, the plurality of connection portions **23R2**, and the plurality of connection portions **23RG** are formed on the first surface of the drive substrate **11**, and the first electrode **13** for forming the sub-pixel **101B** is exposed. Hereinafter, a block including the plurality of light emitting elements **104G**, the plurality of light emitting elements **104R**, the plurality of connection portions **23G1**, the plurality of connection portions **23G2**, the plurality of connection portions **23R1**, the plurality of connection portions **23R2**, the plurality of connection portions **23RG**, and the protective layer **61** formed on the first surface of the light emitting elements **104G** and **104R** and the connection portions **23G1**, **23G2**, **23R1**, **23R2**, and **23RG** is referred to as a multilayer body **105RG**.

(Formation Step of Sidewall)

[0293] Next, as illustrated in FIGS. **45B** and **49B**, the insulating layer **62a** is formed on the first surface of the drive substrate **11** so as to follow the shape of the plurality of multilayer bodies **105RG**, for example, by a CVD method or a vapor deposition method. Next, the insulating layer **62a** is etched back by using, for example, dry etching, so that the sidewall **62** is formed on the side surface of the multilayer body **105RG** and the first electrode **13** for forming the sub-pixel **101B** is exposed again as illustrated in FIGS. **45C** and **49C**.

(Formation Step of Organic Layer Having Blue Color as Emission Color)

[0294] Next, as illustrated in FIGS. **45D** and **49D**, the organic layer **14B** is formed over the entire display region **10A** on the first surface of the drive substrate **11** so as to follow the shape of the multilayer body **105RG** on which the sidewall **62** is formed, for example, by a vapor deposition method.

(Formation Step of Second Electrode)

[0295] Next, as illustrated in FIGS. **45D** and **49D**, the second electrode **15** is formed on the first surface of the organic layer **14B** so as to follow the shape of the multilayer body **105RG** on which the sidewall **62** is formed, for example, by a vapor deposition method or a sputtering method.

(Formation Step of Protective Layer)

[0296] Next, as illustrated in FIGS. **45D** and **49D**, a protective layer **61a** is formed on the first

surface of the second electrode **15** by, for example, a CVD method or a vapor deposition method so as to follow the shape of the multilayer body **105RG** in which the sidewall **62** is formed on the side surface. As a result, each of the plurality of recesses **61b** is formed above the first electrode **13** for forming the sub-pixel **101B**.

(Patterning Step of Organic Layer, Second Electrode and Protective Layer)

[0297] Next, as illustrated in FIGS. **46A** and **50A**, a resist layer **74** is formed in each recess **61b**. Subsequently, as illustrated in FIGS. **46B** and **50B**, the protective layer **61a**, the second electrode **15**, and the organic layer **14B** are processed by, for example, dry etching. As a result, the protective layer **61a**, the second electrode **15**, and the organic layer **14B** located on the multilayer body **105RG** are removed, and the protective layer **61a**, the second electrode **15**, and the organic layer **14B** located between the multilayer body **105RG** and the first electrode **13** for forming the sub-pixel **101B** in plan view are removed. Thereafter, the photoresist layer **73** is removed. As a result, the plurality of light emitting elements **104B** is further formed on the first surface of the drive substrate **11**. Hereinafter, a block including the light emitting element **104B** and the protective layer **61** formed on the first surface of the light emitting element **104B** is referred to as a multilayer body **105B**.

(Formation Step of Sidewall)

[0298] Next, as illustrated in FIGS. **46C** and **50C**, the insulating layer **62a** is formed on the first surface of the drive substrate **11** so as to follow the shapes of the multilayer body **105RG** and the plurality of multilayer bodies **105B**, for example, by a CVD method or a vapor deposition method. Next, the insulating layer **62a** is etched back using, for example, dry etching to form the sidewall **62** on the side surface of each multilayer body **105B** as illustrated in FIGS. **46D** and **50D**.

(Formation Step of Auxiliary Electrode)

[0299] Next, the hole **611** is formed in the protective layer **61** by patterning the protective layer **61** of the multilayer body **105B** by, for example, a photolithography method. More specifically, as illustrated in FIGS. **47A** and **51A**, a photoresist layer **75** having a predetermined pattern is formed on the first surface of the protective layer **61** and the sidewall **62**. Subsequently, as illustrated in FIGS. **47B** and **51B**, the protective layer **61** is processed by dry etching, for example, to form the hole **611** in the protective layer **61**. After the photoresist layer **75** is removed, as illustrated in FIGS. **47C** and **51C**, the auxiliary electrode **63** is formed on the first surface of the protective layer **61** by, for example, a vapor deposition method or a sputtering method, and the connection portion **631** is formed in the hole **611**. As a result, the auxiliary electrode **63** is connected to the second electrode **15** of the light emitting element **104B** via the connection portion **631**.

(Formation Step of Protective Layer)

[0300] Next, as illustrated in FIGS. **47D** and **51D**, a protective layer **64** is formed on the second surface of the auxiliary electrode **63** by, for example, a CVD method or a vapor deposition method. As described above, the intended display device **10** is obtained.

2-2-2 Operation and Effect

[0301] As a method of separately applying an organic layer having a red emission color, an organic layer having a green emission color, and an organic layer having a blue emission color, a pattern deposition method using a fine metal mask (FMM) or a membrane mask, and a method of dissolving a material for forming an organic layer in a solvent and applying each color by inkjet are generally used. However, in these methods, deposition accuracy and coating accuracy are insufficient, and pattern formation may be difficult. In particular, in a high-definition display device of 3000 ppi or more, deposition accuracy and coating accuracy are insufficient, and pattern formation tends to be difficult. Therefore, a method of separately forming each light emitting layer by separately patterning an organic layer having a red emission color, an organic layer having a green emission color, and a light emitting layer having a blue light emission color by a photolithography method has been studied. However, in this method, a total of three photolithography processes are required to separately form each light emitting layer. Therefore,

since the number of manufacturing processes of the display device increases, the throughput may decrease.

[0302] On the other hand, in the method of manufacturing the display device **10** according to the fifth embodiment described above, the organic layer **14R** having a red emission color and the organic layer **14G** having a green emission color are simultaneously patterned by a photolithography method, and then the organic layer **14B** having a blue emission color is separately patterned by a photolithography method, thereby separately forming the organic layers **14R**, **14G**, and **14B**. Therefore, in this method, the organic layers **14R**, **14G**, and **14B** can be separately formed by a total of two photolithography processes. Therefore, since the number of manufacturing processes of the display device **10** can be reduced, the throughput can be improved.

[0303] Furthermore, in the method of manufacturing the display device **10** according to the fifth embodiment described above, the plurality of light emitting elements **12R**, **12G**, and **12B** is formed on the first surface of the drive substrate **11** as follows. First, an area for forming the organic layer **14B** is used as a margin area, and the organic layers **14G** of the two pixel columns LG and the organic layers **14R** of the two pixel columns LR are formed via the masks **71** and **72** by a vapor deposition method or the like. Next, the second electrode **15** and the protective layer **61** are sequentially formed so as to cover the organic layer **14R** and the organic layer **14G**. Next, the organic layer **14R** and the organic layer **14G** are simultaneously patterned by a photolithography method together with the second electrode **15** and the protective layer **61**, thereby forming the plurality of light emitting elements **104R** and **104G** on the first surface of the drive substrate **11**. Next, the organic layer **14B** is formed over the entire display region **10A** by a vapor deposition method or the like, and then the second electrode **15** and the protective layer **61** are sequentially formed on the organic layer **14B**. Next, by patterning the organic layer **14B** together with the second electrode **15** and the protective layer **61** by a photolithography method, a plurality of light emitting elements **104B** is further formed on the first surface of the drive substrate **11**. Therefore, the throughput can be improved, and the definition of the display device **10** can be improved as compared with the case where the organic layers **14R**, **14G**, and **14B** are each formed via a mask by a vapor deposition method. For example, the high-definition display device **10** of 3000 ppi or more can be provided.

[0304] Furthermore, in the method of manufacturing the display device **10** according to the fifth embodiment described above, since the two sub-pixels **101G** and **101G** having the same emission color are adjacent in the X direction, after one organic layer **14G** is formed so as to cover the two adjacent first electrodes **13**, the organic layer **14G** can be separated for each sub-pixel **101G** by a photolithography method. Similarly, since the two sub-pixels **101R** and **101R** having the same emission color are adjacent to each other in the X direction, after one organic layer **14R** is formed so as to cover the two adjacent first electrodes **13**, the organic layer **14R** can be separated for each sub-pixel **101R** by a photolithography method. Therefore, the accuracy of the masks **71** and **72** can be relaxed, and high definition of the display device **10** can be realized.

2-2-3 Modification of Manufacturing Method

[0305] In the above description of the method of manufacturing the display device, an example in which the plurality of organic layers **14G** and the plurality of organic layers **14R** are formed by the vapor deposition method using the mask **71** and the mask **72** has been described, but the method of forming these layers is not limited to this example. For example, the plurality of organic layers **14G** and the plurality of organic layers **14R** may be formed by applying a material for forming the organic layer **14G** and a material for forming the organic layer **14R** on the first surface of the drive substrate **11** by an inkjet method and curing the materials.

3 Example of Case where Display Device Includes Wavelength Selection Unit

[0306] In each example of the display device and the method of manufacturing the display device described above, the description of the wavelength selection unit and the lens exemplified by the color filter and the like is omitted, but this does not deny that the wavelength selection unit and the

lens are provided in the display device of the present disclosure, and the wavelength selection unit and the lens may be provided. Note that the wavelength selection unit is not limited to the color filter.

(Color Filter, Lens Member)

[0307] In the display device **10**, as illustrated in FIG. **26**, a wavelength selection unit and a lens member may be provided on the first surface side of the low refractive index layer **18** in each sub-pixel **101**. As illustrated in FIG. **26**, for example, a color filter **19** can be exemplified as the wavelength selection unit. As the color filter **19**, a filter corresponding to the color type of the sub-pixel **101** is preferably provided. For example, a red filter **19R**, a green filter **19G**, and a blue filter **19B** may be provided as color filters for the sub-pixels **101R**, **101G**, and **101B**, respectively. The color purity can be improved by providing the color filter. At this time, a light absorbing layer **21** is preferably provided between the adjacent color filters **19**. The light absorbing layer **21** can be exemplified by a black matrix portion or the like. Furthermore, a lens unit **20** may be formed on the color filter **19**. Examples of the lens unit **20** include a convex lens. Since the lens unit **20** is formed, a direction in which light travels can be adjusted.

(Relationship Among Normal Lines Passing Through Centers of Light Emitting Unit, Lens Member, and Wavelength Selection Unit)

[0308] In the description below, the relationship among a normal line LN extending through the center of a light emitting unit, a normal line LN' extending through the center of a lens member, and a normal line LN'' extending through the center of a wavelength selection unit is described. Here, the light emitting unit is, for example, the light emitting element **104**. The lens member is, for example, the lens unit **20** provided on the color filter. The wavelength selection unit is, for example, a red filter **19R**, a green filter **19G**, and a blue filter **19B**.

[0309] Note that the size of the wavelength selection units may be changed as appropriate in accordance with light emitted from the light emitting units, or, in a case where the light absorbing units (black matrix portions, for example) are provided between the wavelength selection units of adjacent light emitting units, the size of the light absorbing units may be changed as appropriate in accordance with light emitted from the light emitting units. Furthermore, the size of each wavelength selection unit may be changed as appropriate in accordance with the distance (offset amount) $d_{sub.0}$ between the normal line extending through the center of the light emitting unit and the normal line extending through the center of the wavelength selection unit. The planar shape of each wavelength selection unit may be the same as, similar to, or different from the planar shape of each lens member.

[0310] Hereinafter, with reference to FIGS. **22A**, **22B**, **22C**, and **23**, a relationship of a normal line passing through the center of each part in a case where the light emitting unit **51**, the wavelength selection unit **52**, and the lens member **53** are disposed in this order will be described.

[0311] As illustrated in FIG. **22A**, a normal line LN passing through the center of the light emitting unit **51**, a normal line LN'' passing through the center of the wavelength selection unit **52**, and a normal line LN' passing through the center of the lens member **53** may coincide with each other. That is, $D_{sub.0}=0$ and $d_{sub.0}=0$ may be satisfied. Here, $D_{sub.0}$ represents the distance (offset amount) between the normal line LN extending through the center of the light emitting unit **51** and the normal line LN' extending through the center of the lens member **53**, and $d_{sub.0}$ represents the distance (offset amount) between the normal line LN extending through the center of the light emitting unit **51** and the normal line LN'' extending through the center of the wavelength selection unit **52**.

[0312] As illustrated in FIG. **22B**, the normal line LN extending through the center of the light emitting unit **51** and the normal line LN'' extending through the center of the wavelength selection unit **52** may coincide with each other, but the normal line LN extending through the center of the light emitting unit **51** and the normal line LN'' extending through the center of the wavelength selection unit **52** may not coincide with the normal line LN' extending through the center of the

lens member 53. That is, $D_{\text{sub.0}} > 0$ and $d_{\text{sub.0}} = 0$ may be satisfied.

[0313] As illustrated in FIG. 22C, the normal line LN extending through the center of the light emitting unit 51 may not coincide with the normal line LN'' extending through the center of the wavelength selection unit 52 and the normal line LN' extending through the center of the lens member 53, and the normal line LN'' extending through the center of the wavelength selection unit 52 may coincide with the normal line LN' extending through the center of the lens member 53. That is, $D_{\text{sub.0}} > 0$, $d_{\text{sub.0}} > 0$, and $D_{\text{sub.0}} = d_{\text{sub.0}}$ may be satisfied.

[0314] As illustrated in FIG. 23, the normal line LN passing through the center of the light emitting unit 51, the normal line LN'' passing through the center of the wavelength selection unit 52, and the normal line LN' passing through the center of the lens member 53 may not coincide with each other. That is, $D_{\text{sub.0}} > 0$, $d_{\text{sub.0}} > 0$, and $D_{\text{sub.0}} \neq 0$ may be satisfied. Here, the center of the wavelength selection unit 52 (a position indicated by a black square in FIG. 23) is preferably located on a straight line LL connecting the center of the light emitting unit 51 and the center of the lens member 53 (a position indicated by a black circle in FIG. 23). Specifically, when a distance in the thickness direction (in FIG. 23, the vertical direction) between the center of the light emitting unit 51 and the center of the wavelength selection unit 52 is LL.sub.1, and a distance in the thickness direction between the center of the wavelength selection unit 52 and the center of the lens member 53 is LL.sub.2,

$D_{\text{sub.0}} > d_{\text{sub.0}} > 0$, and [0315] considering manufacturing variations,

[00001] $d_0 : D_0 = LL_1 : (LL_1 + LL_2)$ [0316] is preferably satisfied.

[0317] Here, the thickness direction indicates the thickness direction of the light emitting unit 51, the wavelength selection unit 52, and the lens member 53.

[0318] Hereinafter, with reference to FIGS. 24A, 24B, and 25, a relationship of a normal line passing through the center of each part in a case where the light emitting unit 51, the lens member 53, and the wavelength selection unit 52 are disposed in this order will be described.

[0319] As illustrated in FIG. 24A, a normal line LN passing through the center of the light emitting unit 51, a normal line LN'' passing through the center of the wavelength selection unit 52, and a normal line LN' passing through the center of the lens member 53 may coincide with each other. That is, $D_{\text{sub.0}} > 0$ and $d_{\text{sub.0}} = 0$ may be satisfied.

[0320] As illustrated in FIG. 24B, a normal line LN passing through the center of the light emitting unit 51, a normal line LN'' passing through the center of the wavelength selection unit 52 and the normal line LN' passing through the center of the lens member 53 do not coincide with each other, and the normal line LN'' passing through the center of the wavelength selection unit 52, and a normal line LN' passing through the center of the lens member 53 may coincide with each other. That is, $D_{\text{sub.0}} > 0$, $d_{\text{sub.0}} > 0$, and $D_{\text{sub.0}} = d_{\text{sub.0}}$ may be satisfied.

[0321] As illustrated in FIG. 25, the normal line LN passing through the center of the light emitting unit 51, the normal line LN'' passing through the center of the wavelength selection unit 52, and the normal line LN' passing through the center of the lens member 53 may not coincide with each other. Here, the center of the lens member 53 (the position indicated by a black circle in FIG. 41) is preferably located on the straight line LL connecting the center of the light emitting unit 51 and the center of the wavelength selection unit 52 (the position indicated by a black square in FIG. 41). Specifically, where the distance in the thickness direction (the vertical direction in FIG. 41) between the center of the light emitting unit 51 and the center of the lens member 53 is represented by LL.sub.2, and the distance in the thickness direction between the center of the lens member 53 and the center of the wavelength selection unit 52 is represented by LL.sub.1, [0322] the following expression is preferably satisfied,

$d_{\text{sub.0}} > D_{\text{sub.0}} > 0$ [0323] and, with manufacturing variations being taken into consideration, the following expression is preferably satisfied,

[00002] $D_0:d_0 = LL_2:(LL_1 + LL_2)$

[0324] Here, the thickness direction indicates the thickness direction of the light emitting unit **51**, the wavelength selection unit **52**, and the lens member **53**.

3 Application Example

(Electronic Device)

[0325] The display device **10** according to the first to fifth embodiments described above may be provided in various electronic devices. Especially, the display device is preferably provided in an apparatus requiring high resolution of an image and used near the eyes for viewing in a magnified state, the apparatus including an electronic viewfinder of a video camera or a single-lens reflex camera, a head mounted display, or the like.

Specific Example 1

[0326] FIG. **27A** is a front view illustrating an example of an external appearance of a digital still camera **310**. FIG. **27B** is a rear view illustrating an example of an external appearance of the digital still camera **310**. The digital still camera **310** is of a lens interchangeable single-lens reflex type, and includes an interchangeable imaging lens unit (interchangeable lens) **312** substantially at the center on the front surface of a camera main body (camera body) **311**, and a grip **313** to be held by the photographer on the front left side.

[0327] A monitor **314** is provided at a position shifted to the left side from the center of the rear surface of the camera main body **311**. An electronic viewfinder (eyepiece window) **315** is provided above the monitor **314**. By looking through the electronic viewfinder **315**, the photographer can visually recognize an optical image of the subject guided from the imaging lens unit **312**, and determine a picture composition. As the electronic view finder **315**, any of the display devices **10** according to the above-described first to fifth embodiments and modifications can be used.

Specific Example 2

[0328] FIG. **28** is a perspective view illustrating an example of an external appearance of a head-mounted display **320**. The head-mounted display **320** includes ear hooking portions **322** to be worn on the head of the user on both sides of a display unit **321** in the shape of eyeglasses, for example. As the display unit **321**, any one of the display devices **10** according to the above-described first to fifth embodiments and modifications can be used.

Specific Example 3

[0329] FIG. **29** is a perspective view illustrating an example of an external appearance of a television device **330**. The television device **330** includes, for example, a video display screen unit **331** including a front panel **332** and a filter glass **333**, and the video display screen unit **331** includes any one of the display devices **10** according to the first to fifth embodiments and the modifications described above.

Specific Example 4

[0330] FIG. **30** illustrates an example of an appearance of a see-through head-mounted display **340**. The see-through head-mounted display **340** includes a main body **341**, an arm **342**, and a lens barrel **343**.

[0331] The main body **341** is connected to the arm **342** and eyeglasses **350**. Specifically, an end portion of the main body **341** in the long side direction is coupled to the arm **342**, and one side of a side surface of the main body **341** is coupled to the eyeglasses **350** via a connecting member. Note that the main body **341** may be directly mounted on the head of the human body.

[0332] The main body **341** incorporates a control board for controlling operation of the see-through head-mounted display **340**, and a display unit. The arm **342** connects the main body **341** and the lens barrel **343**, and supports the lens barrel **343**. Specifically, the arm **342** is coupled to an end portion of the main body **341** and an end portion of the lens barrel **343**, and secures the lens barrel **343**. Furthermore, the arm **342** incorporates a signal line for communicating data related to an image to be provided from the main body **341** to the lens barrel **343**.

[0333] The lens barrel **343** projects image light provided from the main body **341** via the arm **342**

toward the eyes of the user wearing the see-through head-mounted display **340** through an eyeglass **351**. In this see-through head-mounted display **340**, the display unit of the main body **341** includes one of the above display devices **10** and the like.

Specific Example 5

[0334] FIG. **31** is a perspective view illustrating an example of an external appearance of a smartphone **360**. As illustrated in FIG. **31**, the smartphone **360** includes a display unit **361** that displays information such as pixels, and an operation unit **362** including a button or the like that receives an operation input by the user. The display device **10** according to the above-described first to fifth embodiments and modifications can be applied to the display unit **361**.

Specific Example 6

[0335] Any of the display devices **10** and the like described above may be included in a vehicle or in various kinds of displays.

[0336] FIGS. **32A** and **32B** are diagrams illustrating an example of an internal configuration of a vehicle **500** provided with various displays. Specifically, FIG. **32A** is a diagram illustrating an example of an internal state of the vehicle **500** from the rear to the front of the vehicle **500**, and FIG. **32B** is a diagram illustrating an example of an internal state of the vehicle **500** from the oblique rear to the oblique front of the vehicle **500**.

[0337] The vehicle **500** includes a center display **501**, a console display **502**, a head-up display **503**, a digital rearview mirror **504**, a steering wheel display **505**, and a rear entertainment display **506**. At least one of these displays includes any one of the above display devices **10** and the like. For example, all of these displays may include one of the above display devices **10** and the like.

[0338] The center display **501** is disposed on the dashboard at a location facing a driver's seat **508** and a passenger seat **509**. FIGS. **32A** and **32B** illustrate an example of the center display **501** having a horizontally long shape extending from the driver's seat **508** side to the passenger seat **509** side, but the screen size and the arrangement place of the center display **501** are arbitrary. The center display **501** can display information sensed by various sensors. As a specific example, the center display **501** can display an image captured by an image sensor, an image of the distance to an obstacle in front of or on a side of the vehicle **500**, the distance being measured by a ToF sensor, a passenger's body temperature detected by an infrared sensor, and the like. The center display **501** can be used to display at least one piece of safety-related information, operation-related information, lifelogs, health-related information, authentication/identification-related information, or entertainment-related information, for example.

[0339] The safety-related information is information about doze sensing, looking-away sensing, sensing of mischief of a child riding together, presence or absence of wearing of a seat belt, sensing of leaving of an occupant, and the like, and is information sensed by a sensor disposed to overlap with the back surface side of the center display **501**, for example. The operation-related information senses a gesture related to an operation performed by an occupant, using a sensor. Gestures to be sensed may include an operation of various kinds of equipment in the vehicle **500**. For example, operations of air conditioning equipment, a navigation device, an audiovisual (AV) device, an illuminating device, and the like are detected. The lifelogs include lifelogs of all the occupants. For example, the lifelogs include an action record of each occupant in the vehicle. By acquiring and storing the lifelogs, it is possible to check the state of each occupant at the time of an accident. The health-related information senses the body temperature of an occupant, using a sensor such as a temperature sensor, and estimates the health condition of the occupant on the basis of the sensed body temperature. Alternatively, the face of the occupant may be imaged with an image sensor, and the health condition of the occupant may be estimated from the imaged facial expression. Moreover, a conversation may be made with an occupant in automatic voice, and the health condition of the occupant may be estimated on the basis of the contents of a response from the occupant. The authentication/identification-related information includes a keyless entry function of performing face authentication using a sensor, and a function of automatically adjusting a seat

height and position through face identification. The entertainment-related information includes a function of detecting, with a sensor, operation information about an AV device being used by an occupant, and a function of recognizing the face of the occupant with sensor and providing content suitable for the occupant through the AV device.

[0340] The console display **502** can be used to display lifelog information, for example. The console display **502** is disposed near a shift lever **511** of a center console **510** between the driver's seat **508** and the passenger seat **509**. The console display **502** can also display information detected by various sensors. Furthermore, the console display **502** may display an image of the surroundings of the vehicle captured with an image sensor, or may display an image of the distance to an obstacle present in the surroundings of the vehicle.

[0341] The head-up display **503** is virtually displayed behind a windshield **512** in front of the driver's seat **508**. The head-up display **503** can be used to display at least one piece of the safety-related information, the operation-related information, the lifelogs, the health-related information, the authentication/identification-related information, or the entertainment-related information, for example. Being virtually disposed in front of the driver's seat **508** in many cases, the head-up display **503** is suitable for displaying information directly related to operations of the vehicle **500**, such as the speed, the remaining amount of fuel (battery), and the like of the vehicle **500**.

[0342] The digital rearview mirror **504** can not only display the rear of the vehicle **500** but also display the state of an occupant in the rear seat, and thus, can be used to display the lifelog information by disposing a sensor on the back surface side of the digital rearview mirror **504** in an overlapping manner, for example.

[0343] The steering wheel display **505** is disposed near the center of a steering wheel **513** of the vehicle **500**. The steering wheel display **505** can be used to display at least one piece of the safety-related information, the operation-related information, the lifelogs, the health-related information, the authentication/identification-related information, or the entertainment-related information, for example. In particular, being located close to the driver's hands, the steering wheel display **505** is suitable for displaying the lifelog information such as the body temperature of the driver, or for displaying information regarding operations of the AV device, the air conditioning equipment, or the like.

[0344] The rear entertainment display **506** is attached to the back side of the driver's seat **508** or the passenger seat **509**, and is for an occupant in the rear seat to enjoy viewing/listening. The rear entertainment display **506** can be used to display at least one piece of the safety-related information, the operation-related information, the lifelogs, the health-related information, the authentication/identification-related information, or the entertainment-related information, for example. In particular, as the rear entertainment display **506** is located in front of an occupant in the rear seat, information related to the occupant in the rear seat is displayed. For example, information regarding an operation of the AV device or the air conditioning equipment may be displayed, or a result of measurement of the body temperature or the like of an occupant in the rear seat with a temperature sensor may be displayed.

[0345] A sensor may be disposed on the back surface side of a display device **10** or the like in an overlapping manner, so that the distance to an object present in the surroundings can be measured in the configuration. Optical distance measurement methods are roughly classified into a passive type and an active type. By a method of the passive type, distance measurement is performed by receiving light from an object, without projecting light from a sensor to the object. Methods of the passive type include a lens focus method, a stereo method, and a monocular vision method. Methods of the active type include distance measurement that is performed by projecting light onto an object, and receiving reflected light from the object with a sensor to measure the distance. Methods of the active type include an optical radar method, an active stereo method, an illuminance difference stereo method, a moire topography method, and an interference method. Any of the display devices **10** and the like described above can be used in distance measurement by

any of these methods. With a sensor disposed on the back surface side of the above display device **10** or the like in an overlapping manner, distance measurement of the passive type or the active type described above can be performed.

[0346] Although the display device according to the first to fourth embodiments, the display device according to the fifth embodiment, the display device according to each example, the method of manufacturing the display device, and the application example have been specifically described as an example of the light emitting device of the present disclosure, the present disclosure is not limited to the display device according to the first to fourth embodiments, the display device according to the fifth embodiment, the display device according to each example, the method of manufacturing the display device, and the application example described above, and various modifications based on the technical idea of the present disclosure can be made.

[0347] For example, the configurations, methods, steps, shapes, materials, numerical values, and the like exemplified in the display device according to the above-described first to fourth embodiments, the display device according to the fifth embodiment, and the display devices according to the respective examples, the method of manufacturing the display device, and the application examples are merely examples, and different configurations, methods, steps, shapes, materials, numerical values, and the like may be used as necessary.

[0348] The above-described display device according to the first to fourth embodiments, the display device according to the fifth embodiment, the display device according to each example, the method of manufacturing the display device, and the configurations, methods, steps, shapes, materials, numerical values, and the like of application examples can be combined with each other without departing from the gist of the present disclosure.

[0349] The materials exemplified in the display device according to the above-described first to fourth embodiments, the display device according to the fifth embodiment, the display device according to each example, the method of manufacturing the display device, and the application examples can be used alone or in combination of two or more unless otherwise specified.

[0350] Furthermore, the present disclosure can also adopt the following configurations.

(1)

[0351] A light emitting device including: [0352] a plurality of sub-pixels that is two-dimensionally arranged and respectively corresponds to a plurality of emission colors; [0353] a connection portion that connects the plurality of different sub-pixels; [0354] a first electrode; and [0355] on an upper side of the first electrode, an organic layer including a light emitting layer, and a second electrode in this order, [0356] in which the first electrode and the organic layer are formed in at least each of the plurality of sub-pixels, [0357] the second electrode is formed in each of the plurality of sub-pixels and the connection portion, [0358] the connection portion is, in a case where a region between the plurality of sub-pixels is defined as an inter-sub-pixel region, formed in a part of the inter-sub-pixel region, and [0359] at least a part of the connection portion connects the plurality of sub-pixels having different emission colors.

(2)

[0360] The light emitting device according to (1) described above, in which [0361] the organic layer includes a plurality of functional layers as layers excluding the light emitting layer, and [0362] at least a part of the functional layers is formed in the sub-pixels and the connection portion.

(3)

[0363] The light emitting device according to (1) or (2) described above, in which [0364] the light emitting layer extends from the sub-pixels to a part of the connection portion.

(4)

[0365] The light emitting device according to any one of (1) to (3) described above, in which [0366] a plurality of the light emitting layers extends to at least some of the connection portions.

(5)

[0367] The light emitting device according to any one of (1) to (4) described above, in which

[0368] the organic layer includes the light emitting layer and a plurality of functional layers as layers excluding the light emitting layer, [0369] in at least a part of the sub-pixels, the light emitting layer and the functional layers include side end surfaces, and [0370] the side end surface of the light emitting layer and the side end surfaces of the plurality of functional layers are aligned.

(6)

[0371] The light emitting device according to any one of (1) to (5) described above, in which [0372] in at least a part of the sub-pixels, each of the organic layer and the second electrode includes a side end surface, and the side end surface of the organic layer and the side end surface of the second electrode are aligned.

(7)

[0373] The light emitting device according to any one of (1) to (6) described above, further including [0374] a protective layer that covers the second electrode, [0375] in which in at least a part of the sub-pixels, each of the organic layer, the second electrode, and the protective layer includes a side end surface, and the side end surface of the organic layer, the side end surface of the second electrode, and the side end surface of the protective layer are aligned.

(8)

[0376] The light emitting device according to any one of (1) to (7) described above, in which [0377] at least some of the connection portions connect the sub-pixels having the same emission color.

(9)

[0378] The light emitting device according to any one of (1) to (8) described above, in which [0379] at least some of the connection portions connect three or more of the sub-pixels.

(10)

[0380] The light emitting device according to any one of (1) to (9) described above, in which [0381] the plurality of sub-pixels is arranged in a layout selected from a delta type, a square type, and a stripe type.

(11)

[0382] The light emitting device according to any one of (1) to (10) described above, in which [0383] two of the sub-pixels having a predetermined color are arranged adjacent to each other.

(12)

[0384] The light emitting device according to (11), in which [0385] the connection portion includes a connection portion that connects the sub-pixels having the same emission color and a connection portion that connects the sub-pixels having different emission colors.

(13)

[0386] The light emitting device according to (1), in which [0387] the plurality of sub-pixels includes a plurality of first sub-pixels having a first emission color, a plurality of second sub-pixels having a second emission color, and a plurality of third sub-pixels having a third emission color, [0388] the plurality of first sub-pixels is arranged such that two of the first sub-pixels are adjacent to each other in a predetermined direction, [0389] the plurality of second sub-pixels is arranged such that two of the second sub-pixels are adjacent to each other in the predetermined direction, and [0390] the plurality of third sub-pixels is arranged between the first sub-pixels and the second sub-pixels.

(14)

[0391] The light emitting device according to (13), in which [0392] the connection portion includes: [0393] a first connection portion that connects the first sub-pixels adjacent to each other; [0394] a second connection portion that connects the second sub-pixels adjacent to each other; and [0395] a third connection portion that connects the first sub-pixels and the second sub-pixels adjacent to each other.

(15)

[0396] The light emitting device according to (14), further including [0397] an auxiliary electrode

provided above the second electrode, [0398] in which the auxiliary electrode is connected to the third sub-pixels.

(16)

[0399] An electronic device including [0400] the display device according to any one of (1) to (10) and (11) to (15) described above.

(17)

[0401] A method of manufacturing a light emitting device, the method including: [0402] a first step of patterning an organic layer including a light emitting layer on a first electrode using a mask determined according to a layout of a plurality of sub-pixels; [0403] a second step of layering a second electrode on the organic layer; and [0404] a third step of, using etching, removing a portion of the organic layer and the second electrode, the portion being out of a combined portion of the sub-pixels and a connection portion that connects the plurality of different sub-pixels.

(18)

[0405] The method of manufacturing a light emitting device according to (17) described above, in which [0406] the mask is determined for each of color type of the plurality of sub-pixels respectively corresponding to each of a plurality of emission colors, and [0407] the first step includes forming the light emitting layer corresponding to the plurality of sub-pixels by changing the mask for each of the color types of the sub-pixels.

(19)

[0408] A method of manufacturing a light emitting device, the method including the steps of: [0409] forming a first organic layer including a first light emitting layer on a first electrode that forms two first sub-pixels adjacent to each other in a predetermined direction via a first mask; [0410] after disposing a second mask such that a first electrode that forms one third sub-pixel is present between an opening of the first mask and an opening of the second mask, forming a second organic layer including a second light emitting layer on a first electrode that forms two second sub-pixels adjacent to each other in the predetermined direction via the second mask; [0411] sequentially forming a second electrode that forms a first sub-pixel and a second sub-pixel, and a first protective layer on the first organic layer and the second organic layer; and [0412] forming the first sub-pixel and the second sub-pixel by patterning the first protective layer, the second electrode, the first organic layer, and the second organic layer such that a connection portion that connects the first sub-pixel and the second sub-pixel remains in an inter-sub-pixel region and the first electrode that forms the third sub-pixel is exposed.

(20)

[0413] The method of manufacturing a light emitting device according to (19), further including the steps of: [0414] sequentially forming a third organic layer including a third light emitting layer, a second electrode that forms a third sub-pixel, and a second protective layer so as to cover the first sub-pixel and the second sub-pixel; and [0415] patterning the third organic layer, the second electrode that forms the third sub-pixel, and the second protective layer to form the third sub-pixel.

REFERENCE SIGNS LIST

[0416] **10** Display device [0417] **10A** Display region [0418] **11** Drive substrate [0419] **11A** Substrate [0420] **11B** Insulating layer [0421] **11C** Wiring [0422] **12** Insulating layer [0423] **12A** Opening [0424] **13** First electrode [0425] **14** Organic layer [0426] **14B** Organic layer [0427] **14B0** Main body [0428] **14G** Organic layer [0429] **14G0** Main body [0430] **14G1** Coupling portion [0431] **14G2** Coupling portion [0432] **14G3** Extension portion [0433] **14GL** Main body [0434] **14R** Organic layer [0435] **14R0** Main body [0436] **14R1** Coupling portion [0437] **14R2** Coupling portion [0438] **14R3** Extension portion [0439] **14RL** Main body [0440] **15** Second electrode [0441] **15M0** Main body [0442] **15M1** Coupling portion [0443] **15M2** Coupling portion [0444] **15ML** Main body [0445] **16** First protective layer [0446] **17** Second protective layer [0447] **18** Low refractive index layer [0448] **19** Color filter [0449] **19B** Blue filter [0450] **19G** Green filter [0451] **19R** Red filter [0452] **20** Lens unit [0453] **21** Light absorbing layer [0454] **22** Multilayer structure

[0455] **23**, **23G1**, **23G2**, **23G3**, **23R1**, **23R2**, **23R3**, **23RG** Connection portion [0456] **24** Side wall [0457] **25** Functional layer [0458] **30** Reflector [0459] **31** Optical adjustment layer [0460] **32** Oxide film [0461] **51** Light emitting unit [0462] **52** Wavelength selection unit [0463] **53** Lens member [0464] **101** Sub-pixel [0465] **101B** Sub-pixel [0466] **101G** Sub-pixel [0467] **101R** Sub-pixel [0468] **104** Light emitting element [0469] **104B** Light emitting element [0470] **104G** Light emitting element [0471] **104R** Light emitting element [0472] **105** H driver [0473] **106** V driver [0474] **107** Control circuit [0475] **122** First portion [0476] **123** Second portion [0477] **125A** First layer [0478] **125B** Second layer [0479] **126** Layer [0480] **140** Hole injection layer [0481] **141** Hole transport layer [0482] **142** Light emitting layer [0483] **142B** Light emitting layer [0484] **142G** Light emitting layer [0485] **142R** Light emitting layer [0486] **143** Electron transport layer [0487] **144** Electron injection layer [0488] **150** Mask [0489] **150B** Mask [0490] **150G** Mask [0491] **150R** Mask [0492] **151** Resist [0493] **241** Side end surface [0494] **242** Side end surface [0495] **243** Side end surface [0496] **310** Digital still camera [0497] **320** Head-mounted display [0498] **330** Television device [0499] **340** See-through head-mounted display [0500] **360** Smartphone [0501] **500** Vehicle [0502] **501** Center display [0503] **502** Console display [0504] **503** Head-up display [0505] **504** Digital rearview mirror [0506] **505** Steering wheel display [0507] **506** Rear entertainment display [0508] A.sub.X Axis [0509] **BK1** First block [0510] **BK2** Second block [0511] **DP** Light emitting surface [0512] **LR** Pixel column [0513] **LG** Pixel column [0514] **LB** Pixel column [0515] **LN** Normal line [0516] **LN'** Normal line [0517] **M** Inter-sub-pixel region [0518] **MU** Section line [0519] **XS** Region

Claims

1. A light emitting device comprising: a plurality of sub-pixels that is two-dimensionally arranged and respectively corresponds to a plurality of emission colors; a connection portion that connects the plurality of different sub-pixels; a first electrode; and on an upper side of the first electrode, an organic layer including a light emitting layer, and a second electrode in this order, wherein the first electrode and the organic layer are formed in at least each of the plurality of sub-pixels, the second electrode is formed in each of the plurality of sub-pixels and the connection portion, the connection portion is, in a case where a region between the plurality of sub-pixels is defined as an inter-sub-pixel region, formed in a part of the inter-sub-pixel region, and at least a part of the connection portion connects the plurality of sub-pixels having different emission colors.
2. The light emitting device according to claim 1, wherein the organic layer includes a plurality of functional layers as layers excluding the light emitting layer, and at least a part of the functional layers is formed in the sub-pixels and the connection portion.
3. The light emitting device according to claim 1, wherein the light emitting layer extends from the sub-pixels to a part of the connection portion.
4. The light emitting device according to claim 1, wherein a plurality of the light emitting layers extends to at least some of the connection portions.
5. The light emitting device according to claim 1, wherein the organic layer includes the light emitting layer and a plurality of functional layers as layers excluding the light emitting layer, in at least a part of the sub-pixels, the light emitting layer and the functional layers include side end surfaces, and the side end surface of the light emitting layer and the side end surfaces of the plurality of functional layers are aligned.
6. The light emitting device according to claim 1, wherein in at least a part of the sub-pixels, each of the organic layer and the second electrode includes a side end surface, and the side end surface of the organic layer and the side end surface of the second electrode are aligned.
7. The light emitting device according to claim 1, further comprising a protective layer that covers the second electrode, wherein in at least a part of the sub-pixels, each of the organic layer, the second electrode, and the protective layer includes a side end surface, and the side end surface of

the organic layer, the side end surface of the second electrode, and the side end surface of the protective layer are aligned.

8. The light emitting device according to claim 1, wherein at least some of the connection portions connect the sub-pixels having a same emission color.

9. The light emitting device according to claim 1, wherein at least some of the connection portions connect three or more of the sub-pixels.

10. The light emitting device according to claim 1, wherein the plurality of sub-pixels is arranged in a layout selected from a delta type, a square type, and a stripe type.

11. The light emitting device according to claim 1, wherein two of the sub-pixels having a predetermined color are arranged adjacent to each other.

12. The light emitting device according to claim 11, wherein the connection portion includes a connection portion that connects the sub-pixels having a same emission color and a connection portion that connects the sub-pixels having different emission colors.

13. The light emitting device according to claim 1, wherein the plurality of sub-pixels includes a plurality of first sub-pixels having a first emission color, a plurality of second sub-pixels having a second emission color, and a plurality of third sub-pixels having a third emission color, the plurality of first sub-pixels is arranged such that two of the first sub-pixels are adjacent to each other in a predetermined direction, the plurality of second sub-pixels is arranged such that two of the second sub-pixels are adjacent to each other in the predetermined direction, and the plurality of third sub-pixels is arranged between the first sub-pixels and the second sub-pixels.

14. The light emitting device according to claim 13, wherein the connection portion includes: a first connection portion that connects the first sub-pixels adjacent to each other; a second connection portion that connects the second sub-pixels adjacent to each other; and a third connection portion that connects the first sub-pixels and the second sub-pixels adjacent to each other.

15. The light emitting device according to claim 14, further comprising an auxiliary electrode provided above the second electrode, wherein the auxiliary electrode is connected to the third sub-pixels.

16. An electronic device comprising the display device according to claim 1.

17. A method of manufacturing a light emitting device, the method comprising: a first step of patterning an organic layer including a light emitting layer on a first electrode using a mask determined according to a layout of a plurality of sub-pixels; a second step of layering a second electrode on the organic layer; and a third step of, using etching, removing a portion of the organic layer and the second electrode, the portion being out of a combined portion of the sub-pixels and a connection portion that connects the plurality of different sub-pixels.

18. The method of manufacturing a light emitting device according to claim 17, wherein the mask is determined for each of color type of the plurality of sub-pixels respectively corresponding to each of a plurality of emission colors, and the first step includes forming the light emitting layer corresponding to the plurality of sub-pixels by changing the mask for each of the color types of the sub-pixels.

19. A method of manufacturing a light emitting device, the method comprising the steps of: forming a first organic layer including a first light emitting layer on a first electrode that forms two first sub-pixels adjacent to each other in a predetermined direction via a first mask; after disposing a second mask such that a first electrode that forms one third sub-pixel is present between an opening of the first mask and an opening of the second mask, forming a second organic layer including a second light emitting layer on a first electrode that forms two second sub-pixels adjacent to each other in the predetermined direction via the second mask; sequentially forming a second electrode that forms a first sub-pixel and a second sub-pixel, and a first protective layer on the first organic layer and the second organic layer; and forming the first sub-pixel and the second sub-pixel by patterning the first protective layer, the second electrode, the first organic layer, and the second organic layer such that a connection portion that connects the first sub-pixel and the

second sub-pixel remains in an inter-sub-pixel region and the first electrode that forms the third sub-pixel is exposed.

20. The method of manufacturing a light emitting device according to claim 19, further comprising the steps of: sequentially forming a third organic layer including a third light emitting layer, a second electrode that forms a third sub-pixel, and a second protective layer so as to cover the first sub-pixel and the second sub-pixel; and patterning the third organic layer, the second electrode that forms the third sub-pixel, and the second protective layer to form the third sub-pixel.
